# The Sounding and Wind Map Pages

## Profiles, indices, time–height sections, radiosonde checks and wind maps from aircraft data

Otso Laakso (OH2GAX)
Tuusula, Finland · near Helsinki-Vantaa airport (EFHK)

**In short.** The Sounding page turns the temperatures and winds that MODE-S Wind derives from aircraft transponder replies into the tools a meteorologist uses for upper-air data. Five buttons choose the view. Area · now, Area · hour and Flight draw one vertical profile on a Skew-T diagram, with an Indices panel of stability indices and characteristic levels and a table of the layer values; the Jokioinen radiosonde can be drawn beside it, and its indices are computed by the same code for a fair comparison. Time–height puts the hourly profiles side by side, so that fronts, inversions, jet streams, sinking air and the daily cycle are seen as they pass over the receiver — and, derived from the same data, the vertical motion of the air and the spin (vorticity) of the wind field. Sonde check measures how well the aircraft profile agrees with the radiosonde, and how quickly a sounding goes out of date. The Wind Map page looks at the same data from above: the wind and temperature of one flight level on a map, with isotachs, streamlines, the jet stream, height contours of the pressure surface and comparisons in time and between levels.
This guide explains each view of the Sounding page (Sections 1–6) and the Wind Map page (Section 7), and the physics behind what they show. How to read the Skew-T diagram itself — the lines, the automatic Analysis, the airport low-level profile — is explained in the companion guide *Reading the Atmosphere Profile (Skew-T)*; here it is only summarised. The examples come from 16–19 October 2026: a cold front on 17 October and a strong high-pressure area (QNH 1040 hPa) behind it.
**New in version 2.2:** a first trial of a *moisture indicator* from the aircraft heights in the time–height section (Section 5.9). **Version 2.1:** tropopause and maximum-wind lines, the radiosonde as its own column and saving the picture in the time–height section (Section 5.10); isotherms, a vertical cross-section along a drawn line and all time-slider frames loaded at once on the Wind Map (Sections 7.4 and 7.9); the GNSS height calibration with per-airframe altimeter offsets in the Sonde check (Section 6). **Version 2.0:** the *heights of the pressure surfaces* from the aircraft's GNSS height, with three new fields in the time–height section — height (D-value), height change and a thickness-temperature check (Section 5.8) — and the *Wind Map* page with its jet-stream symbol and geostrophic wind (Section 7). Version 1.1 added vertical motion and vorticity (Sections 5.6 and 5.7).

## Contents

**1 The page at a glance** **2**

**2 Where the profile comes from** **3**

**3 One profile: Area, hour or flight** **3**
3.1 The Skew-T diagram in brief . . . 3
3.2 The radiosonde beside the aircraft . . . 4
3.3 A single flight . . . 5
3.4 The level table . . . 5

**4 The Indices panel** **6**
4.1 Surface and parcel . . . 6
4.2 Levels . . . 7
4.3 Lapse rates, thickness and wind . . . 7

**5 The time–height section** **8**
5.1 How to read the picture . . . 8
5.2 Temperature anomaly: air masses and fronts . . . 8
5.3 Temperature change over 3 hours: advection as it happens . . . 9
5.4 Wind speed: jets and the thermal wind . . . 10
5.5 Lapse rate: stability and inversions . . . 11
5.6 Vertical motion: rising and sinking air . . . 11
5.7 Vorticity: the spin of the wind field . . . 12
5.8 Heights of the pressure surfaces and thickness temperature . . . 14
5.9 Moisture (trial) . . . 16
5.10 Tropopause, maximum wind, sonde columns and saving the picture . . . 19
5.11 Isotherms and the freezing level . . . 19
5.12 Longer periods . . . 19

5.13 A suggested way to use it . . . . . 20

**6 Sonde check: how good is the aircraft profile? 20**

**7 The Wind Map page 22**
7.1 The page at a glance . . . . . 22
7.2 How the map is made . . . . . 23
7.3 $\Delta T$: warm and cold air at the level . . . . . 23
7.4 Isotachs and streamlines . . . . . 24
7.5 Jet stream . . . . . 26
7.6 Heights and the geostrophic wind . . . . . 28
7.7 Radiosonde at the level . . . . . 29
7.8 Compare: change in time and the thermal wind . . . . . 29
7.9 Vertical cross-section along a line . . . . . 31
7.10 Time slider, animation and settings . . . . . 32
7.11 A suggested way to use it . . . . . 32

**8 Limitations 32**

**9 Quick reference 34**

**A Formulas 34**

# 1 The page at a glance

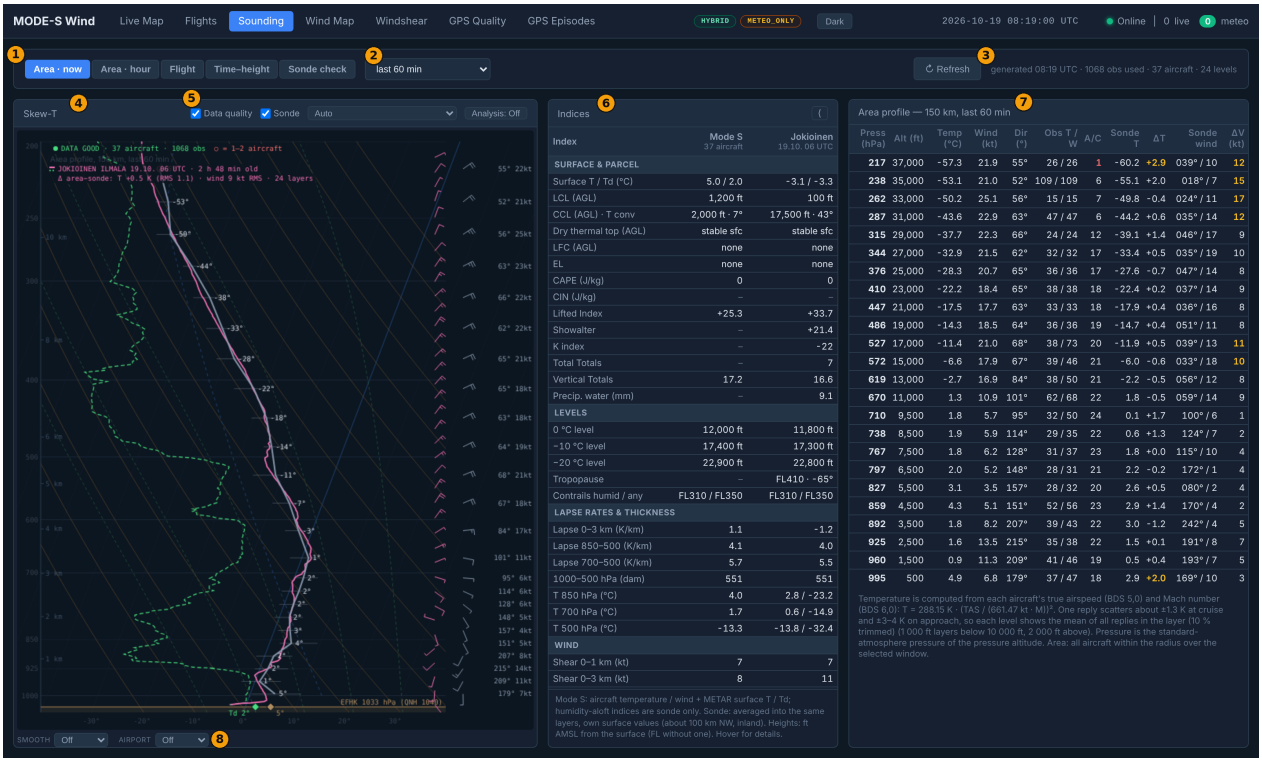



**Figure 1.** The Sounding page, Area · now, 19 October 2026 07:19–08:19 UTC, with Sonde on. (1) view buttons, (2) the controls of the view (here the time window), (3) refresh and summary: observations, aircraft and layers used, (4) Skew-T diagram: grey Mode S area profile, pink radiosonde, green dashed sonde dewpoint, (5) Data quality, Sonde with the sounding selector, Analysis, (6) Indices panel: Mode S and Jokioinen side by side, (7) level table with the sonde columns, (8) Smooth and Airport. Surface values from the METAR stored for 07:50 UTC (5/2 °C, QNH 1040).

The five buttons at the top left ((**1**) in Fig. 1) choose what the page shows; the choice is remembered in the browser.

**Area** · **now** — the live area profile: all aircraft within 150 km of the receiver over the last 30 or 60 minutes, 2 or 3 hours (default 60 minutes), with the current METAR at the ground.

**Area** · **hour** — the *hourly profile archive*: one area profile stored for every completed UTC hour, kept indefinitely, also after the observations themselves have been purged. Pick a date and an hour; the list shows how many observations each

hour has. The METAR of that hour (temperature, dewpoint, QNH, lowest cloud) is stored with it (from 11 October 2026).

**Flight** — the profile of one climbing or descending aircraft, chosen from a list of recent flights with a large enough altitude range (or opened with the Sounding button of the Flights page).

**Time–height** — the hourly profiles side by side over 24 h to 14 days (Section 5).

**Sonde check** — the automatic verification against the radiosonde (Section 6).

The first three views share the same layout: the Skew-T on the left, the Indices panel in the middle and the level table on the right. On narrow screens the table moves below the diagram, and the ⟨ button hides the Indices panel (a slim Indices tab brings it back).

## 2 Where the profile comes from

**Aircraft as weather sensors.** Airliners answer the interrogations of ground radars with *Enhanced Surveillance* reports. Two of them carry the airspeeds and directions of the aircraft: **BDS 5,0** (true track, groundspeed, true airspeed TAS, roll angle) and **BDS 6,0** (magnetic heading, Mach number, indicated airspeed). From each pair MODE-S Wind computes

- the **air temperature**, because the ratio of true airspeed to Mach number is the speed of sound, which depends only on temperature: $T = 288.15\,\left(\mathrm{TAS}/(661.47\,\mathrm{kt}\cdot M)\right)^2\,\mathrm{K}$;
- the **wind**, as the difference between the ground vector (track, groundspeed) and the air vector (true heading, TAS) — the wind triangle. The heading is corrected for magnetic declination and for each airframe's own heading offset, which the system learns.

A single reply pair scatters by about $\pm 1.3$ K at cruise and 3–4 K at approach speeds, because Mach is sent in steps of 0.004 and TAS in steps of 2 kt. Averaging many replies removes most of this; no per-airframe temperature calibration is needed (the README explains why).

**Pressure altitude is pressure.** The altitude an aircraft reports is its *pressure altitude*: the height at which the measured static pressure would occur in the International Standard Atmosphere (ISA), referenced to 1013.25 hPa. FL300 is therefore always the 301 hPa surface, whatever the weather. The page uses this directly as the vertical coordinate: each layer's pressure is the ISA pressure of its pressure altitude. True heights above the sea differ from pressure altitude with the QNH and with the temperature of the air column; the Indices panel computes true heights from the METAR surface pressure (Section 4).

**The area profile.** Observations within 150 km of the receiver are grouped into layers 1 000 ft deep up to 10 000 ft and 2 000 ft deep above (to 46 000 ft). For each layer the **temperature** is the 10 % trimmed mean (the warmest and coldest tenth removed) and the **wind** the vector mean. A daytime hour typically has 25–70 different aircraft, about 20 per layer; around midnight UTC only 2–8 aircraft fly in the area. The profile is thus an *area and time average*: 300 km across and 60 minutes long. A radiosonde is the opposite: one point, measured within about an hour, but at full vertical resolution and with humidity.

**What is not measured.** Aircraft report no humidity. The only moisture value is the METAR dewpoint at the airport, so the Mode S column of the Indices panel cannot give indices that need humidity aloft.

## 3 One profile: Area, hour or flight

### 3.1 The Skew-T diagram in brief

The diagram is the same as the Atmosphere Profile of the Live Map, only larger. Pressure is the vertical axis (logarithmic, 1000 hPa at the bottom), temperature runs along lines tilted 45° to the right. The grey curve is the Mode S temperature with the layer values, the short horizontal bars the 10–90 % range of the replies; wind barbs and their direction and speed are on the right. Faint brown lines are dry adiabats, dashed green lines moist adiabats, the bold blue diagonal is 0 °C and the dashed blue line the ISA. At the bottom the brown ground line is the station pressure from the METAR QNH, with the METAR temperature and dewpoint.

The controls work as on the Live Map and share their settings with it:

- Data quality: a badge with the number of aircraft and observations (`DATA GOOD` $\geq 20$ aircraft per hour of window, `FAIR` 8–19, `LOW` $< 8$), and red rings on layers measured by only one or two aircraft;
- Analysis: Off / Levels / Stability / Turbulence / Advection / Full — tropopause, isotherms, jets, inversions, stability strip, turbulence risk, warm and cold advection, contrail levels, thermal top and surface parcel;
- Smooth: a running mean of 3 or 5 layers on the temperature curve;
- Airport: the low-level profile from the landings and departures at EFHK (200 ft bands) of the shown period — the selected window, that hour, or the flight's time $\pm 15$ minutes.

### 3.2 The radiosonde beside the aircraft

The Finnish Meteorological Institute launches a radiosonde at Jokioinen (about 100 km north-west of the receiver, inland) at about 23:30, 05:30 and 11:30 UTC for the 00, 06 and 12 UTC soundings. MODE-S Wind fetches them from FMI open data as they appear — often while the balloon is still climbing — and stores them. With Sonde ticked the sounding is drawn in pink, its dewpoint as a dashed green curve and its winds as pink barbs, and a label gives its age and the mean difference area − sonde.

The Auto setting of the sounding selector chooses: in Area · now the newest sounding; in Area · hour and Flight the newest one launched by the end of that hour or flight (at most 36 h older), with its age counted to the middle of the hour or flight. Any stored sounding can also be picked by hand, for example the one launched *after* an hour, to see how the atmosphere had changed. Partial soundings show their top (`· to 350 hPa`); FMI's open-data copy of a flight often ends at 250–550 hPa.

The two profiles never agree exactly, and the differences are informative. In calm weather (Fig. 1) they agree within about 1 K and 10 kt through most of the column. When a front lies between Jokioinen and Helsinki they can differ by several degrees (Fig. 2).

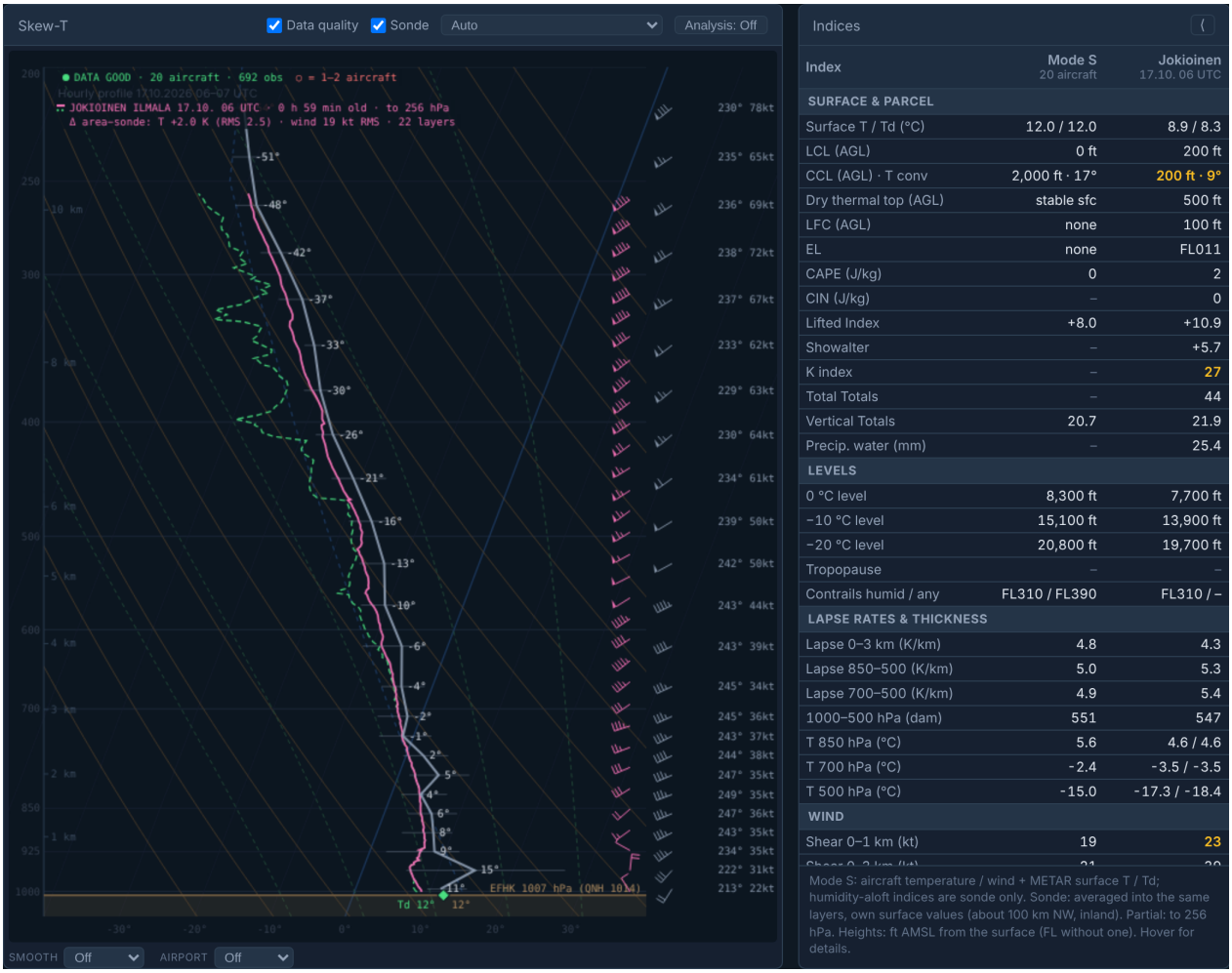



**Figure 2.** Area · hour 17 October 2026 06–07 UTC with the 06 UTC Jokioinen sounding. Over Helsinki the aircraft still measure the warm, south-westerly air ahead of a cold front: 15 °C and 222°/31 kt at 1 500 ft above a cooler, saturated surface layer (METAR 12/12, BKN006). At Jokioinen, already behind the front, the lowest 2 000 ft are 3–8 K colder with a northerly wind (357–006°), while above about 3 000 ft both show the same south-westerly flow. The whole column is warmer over Helsinki: 1000–500 hPa thickness 551 against 547 dam, about 2 K.

This is a textbook picture of a cold front: the cold air advances as a shallow wedge under the warm air, so at a given moment the front is already past at the ground at Jokioinen but not yet over Helsinki, and aloft both sites are still in the same air. The front reached EFHK at about 08–09 UTC (the wind at 1 000 ft turned from 244° to 353°, the METAR temperature fell from 12 to 9 °C). Section 5 shows how the same front looks in the time–height section.

### 3.3 A single flight

Flight draws the replies of one aircraft as it climbs out or descends, with its own wind and temperature layers (Fig. 3). It is a single instrument at a single time — the closest thing to a radiosonde the aircraft data offer — but every layer rests on one to six replies of one airframe, so the zigzags of 2–3 K are mostly Mach and TAS steps rather than real structure. Compare it with the area profile of the same hour before reading anything into a single bend. A flight's profile also has gaps where the radar happened not to interrogate the aircraft; the note under the Indices panel lists them, and values above a gap are less certain. A flight has no METAR, so the surface-based indices are left out.

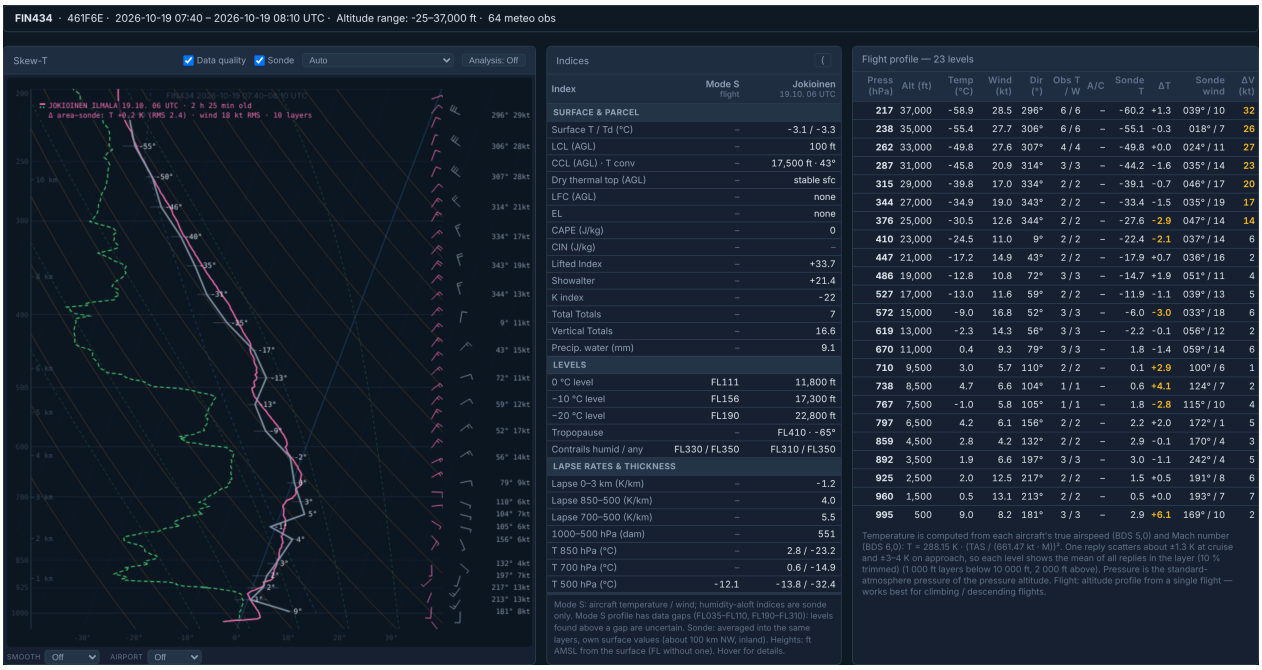
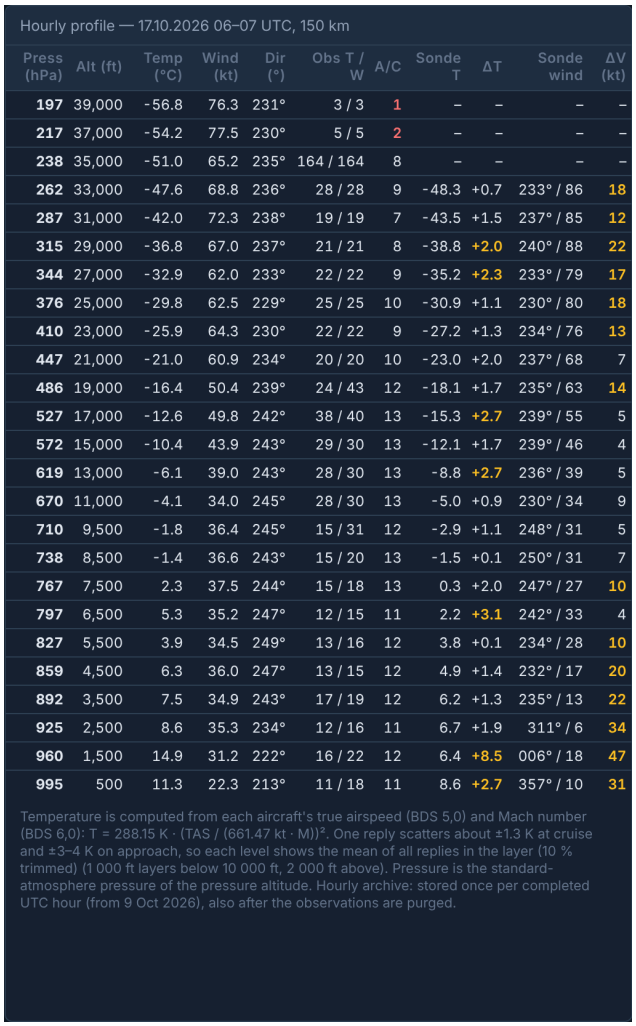



**Figure 3.** Flight: a departure on 19 October 2026 07:40–08:10 UTC with the 06 UTC sounding. Both show the nearly isothermal layer of 0 to +5 °C between about 2 000 and 11 000 ft; the aircraft's single replies zigzag by 2–4 K around it. Above FL250 the aircraft measured north-westerly 20–28 kt where the sonde, 100 km away and two hours earlier, had north-easterly 7–19 kt: under the high the winds were light and variable, so small differences in place and time matter.

### 3.4 The level table

The table lists every layer from the top down, as in the diagram: ISA pressure, layer centre (pressure altitude), temperature (hover: 10–90 % range), wind speed (hover: vector standard deviation) and direction, the number of replies for temperature and wind, and the number of different aircraft (A/C, red when only one or two). With Sonde on, four columns follow: the sonde temperature at the layer's pressure, $\Delta T$ (layer − sonde, amber beyond 2 K), the sonde wind and $\Delta V$, the length of the vector difference (amber beyond 10 kt). In Fig. 4 the front is the column of large $\Delta T$ and $\Delta V$ in the lowest 3 000 ft.



Hourly profile — 17.10.2026 06–07 UTC, 150 km

| Press (hPa) | Alt (ft) | Temp (°C) | Wind (kt) | Dir (°) | Obs T / W | A/C | Sonde T | ΔT | Sonde wind | ΔV (kt) |
|---|---|---|---|---|---|---|---|---|---|---|
| 197 | 39,000 | -56.8 | 76.3 | 231° | 3 / 3 | 1 | – | – | – | – |
| 217 | 37,000 | -54.2 | 77.5 | 230° | 5 / 5 | 2 | – | – | – | – |
| 238 | 35,000 | -51.0 | 65.2 | 235° | 164 / 164 | 8 | – | – | – | – |
| 262 | 33,000 | -47.6 | 68.8 | 236° | 28 / 28 | 9 | -48.3 | +0.7 | 233° / 86 | 18 |
| 287 | 31,000 | -42.0 | 72.3 | 238° | 19 / 19 | 7 | -43.5 | +1.5 | 237° / 85 | 12 |
| 315 | 29,000 | -36.8 | 67.0 | 237° | 21 / 21 | 8 | -38.8 | +2.0 | 240° / 88 | 22 |
| 344 | 27,000 | -32.9 | 62.0 | 233° | 22 / 22 | 9 | -35.2 | +2.3 | 233° / 79 | 17 |
| 376 | 25,000 | -29.8 | 62.5 | 229° | 25 / 25 | 10 | -30.9 | +1.1 | 230° / 80 | 18 |
| 410 | 23,000 | -25.9 | 64.3 | 230° | 22 / 22 | 9 | -27.2 | +1.3 | 234° / 76 | 13 |
| 447 | 21,000 | -21.0 | 60.9 | 234° | 20 / 20 | 10 | -23.0 | +2.0 | 237° / 68 | 7 |
| 486 | 19,000 | -16.4 | 50.4 | 239° | 24 / 43 | 12 | -18.1 | +1.7 | 235° / 63 | 14 |
| 527 | 17,000 | -12.6 | 49.8 | 242° | 38 / 40 | 13 | -15.3 | +2.7 | 239° / 55 | 5 |
| 572 | 15,000 | -10.4 | 43.9 | 243° | 29 / 30 | 13 | -12.1 | +1.7 | 239° / 46 | 4 |
| 619 | 13,000 | -6.1 | 39.0 | 243° | 28 / 30 | 13 | -8.8 | +2.7 | 236° / 39 | 5 |
| 670 | 11,000 | -4.1 | 34.0 | 245° | 28 / 30 | 13 | -5.0 | +0.9 | 230° / 34 | 9 |
| 710 | 9,500 | -1.8 | 36.4 | 245° | 15 / 31 | 12 | -2.9 | +1.1 | 248° / 31 | 5 |
| 738 | 8,500 | -1.4 | 36.6 | 243° | 15 / 20 | 13 | -1.5 | +0.1 | 250° / 31 | 7 |
| 767 | 7,500 | 2.3 | 37.5 | 244° | 15 / 18 | 13 | 0.3 | +2.0 | 247° / 27 | 10 |
| 797 | 6,500 | 5.3 | 35.2 | 247° | 12 / 15 | 11 | 2.2 | +3.1 | 242° / 33 | 4 |
| 827 | 5,500 | 3.9 | 34.5 | 249° | 13 / 16 | 12 | 3.8 | +0.1 | 234° / 28 | 10 |
| 859 | 4,500 | 6.3 | 36.0 | 247° | 13 / 15 | 12 | 4.9 | +1.4 | 232° / 17 | 20 |
| 892 | 3,500 | 7.5 | 34.9 | 243° | 17 / 19 | 12 | 6.2 | +1.3 | 235° / 13 | 22 |
| 925 | 2,500 | 8.6 | 35.3 | 234° | 12 / 16 | 11 | 6.7 | +1.9 | 311° / 6 | 34 |
| 960 | 1,500 | 14.9 | 31.2 | 222° | 16 / 22 | 12 | 6.4 | +8.5 | 006° / 18 | 47 |
| 995 | 500 | 11.3 | 22.3 | 213° | 11 / 18 | 11 | 8.6 | +2.7 | 357° / 10 | 31 |

Temperature is computed from each aircraft's true airspeed (BDS 5,0) and Mach number (BDS 6,0): $T = 288.15\ \mathrm{K} \cdot (\mathrm{TAS} / (661.47\ \mathrm{kt} \cdot M))^2$. One reply scatters about ±1.3 K at cruise and ±3–4 K on approach, so each level shows the mean of all replies in the layer (10 % trimmed) (1 000 ft layers below 10 000 ft, 2 000 ft above). Pressure is the standard-atmosphere pressure of the pressure altitude. Hourly archive: stored once per completed UTC hour (from 9 Oct 2026), also after the observations are purged.

**Figure 4.** Level table of Fig. 2. The lowest layers differ from the sonde by up to +8.5 K and 47 kt (opposite wind directions across the front); higher up the differences are mostly within ±2 K and 5–20 kt. The top three layers are above the end of the sounding.

## 4 The Indices panel

The panel lists stability indices and characteristic levels for the shown profile (Mode S) and, with Sonde on, for the radiosonde (Jokioinen). Both columns are computed by the same code, and the sounding is first averaged into the same 1 000 / 2 000 ft layers as the aircraft profile, so the comparison shows measurement differences, not differences in resolution. The sonde column uses the sonde's own surface values; the Mode S column uses the METAR. Hover a label for its definition and a value for details (its pressure, all inversions or turbulence layers, the further crossings of an isotherm, or why a value cannot be given). Amber and red mark values that indicate instability or strong gradients.

Heights are computed *hypsometrically* from the surface point: each layer's thickness follows from its mean temperature (Section A). Parcel levels are given in feet above the ground (AGL), isotherms in feet above mean sea level; without a surface point (a flight) only flight levels are given. Temperatures are not interpolated across a data gap of more than one missing 2 000 ft layer, and the note lists such gaps.

### 4.1 Surface and parcel

These rows lift the surface air — the METAR temperature and dewpoint — and compare it with the profile. They describe convection that starts from the ground.

**LCL** Lifting condensation level: rising air cools at the dry adiabatic rate (9.8 K/km) until it reaches its dewpoint and condenses. With a well-mixed boundary layer this is the base of cumulus clouds, roughly 400 ft for each degree of spread $T - T_d$. In Fig. 2 the METAR 12/12 gives an LCL at the ground: the air was saturated (low stratus).

**CCL and convective temperature** The convective condensation level is where the line of constant moisture through the surface dewpoint meets the profile; the convective temperature is the surface temperature that would carry dry thermals up to it. When the surface reaches it (amber), cumulus from surface heating alone can form.

**Dry thermal top** The surface temperature lifted along the dry adiabat until it meets the profile: the depth of dry thermals

and of daytime mixing. `stable sfc` means the air above the ground is warmer than surface air lifted to it — a surface inversion.

**LFC, EL, CAPE, CIN** Above the LCL the parcel follows a moist adiabat. If it becomes warmer than its surroundings (level of free convection, LFC) it rises freely until it is again as cold as the environment (equilibrium level, EL, roughly the top of convective clouds). CAPE is the energy the parcel gains in between (amber $\geq 100$, red $\geq 1\,000$ J/kg); CIN is the energy needed to lift it to the LFC.

**Lifted Index, Showalter** Environment minus parcel temperature at 500 hPa for a surface parcel (LI) or an 850 hPa parcel (Showalter, sonde only). Negative values mean unstable air (amber $\leq -1$, red $\leq -5$ for LI).

**K index, Total Totals, Vertical Totals** Classic air-mass thunderstorm indices from the temperatures and dewpoints at 850, 700 and 500 hPa. Only Vertical Totals, $T_{850} - T_{500}$, needs no humidity and is available from the aircraft; values over about 30 mean steep mid-level lapse rates.

**Precipitable water** All the water vapour of the column, as millimetres of liquid water (sonde only).

Under a high-pressure area, as in Fig. 1, all of these say the same thing: very stable air (Lifted Index $+25$ to $+34$, no LFC, `stable sfc`). The LCL of the Mode S column (1 200 ft) and of the sonde (100 ft) differ because the surface values differ: 5/2 °C at the airport against $-3/-3$ °C at the sonde site, inland after a clear night.

### 4.2 Levels

**0, −10, −20 °C levels** The lowest crossing of each isotherm; `(+n)` and the hover list further crossings, which mean a warm or cold layer aloft. The freezing level matters for the type of precipitation and for airframe icing, which is most common in supercooled cloud between 0 and about −15 °C.

**Tropopause** The WMO definition: the lowest level above 5 km where the lapse rate falls to 2 K/km or less and stays so for at least 2 km. The aircraft data often end at about FL410, so a high tropopause is found only by the sonde (FL410 in Fig. 1).

**Contrails** The lowest level where a jet engine's exhaust forms a contrail if the air is saturated / at any humidity (Schmidt–Appleman criterion). Contrails persist only in air that is supersaturated with respect to ice, which the aircraft cannot measure.

**When the 0 °C level is ill-defined.** On 18–19 October the air between the ground and about 11 000 ft was nearly isothermal at 0 to +4 °C, the result of sinking air under the high (Section 5.5). In such a layer the first crossing of 0 °C is decided by a few tenths of a degree, and noise of ±1 K can move it by several thousand feet from one hour to the next (the jumping 0 °C line on 19 October in Figs. 6–9). This is a property of the atmosphere, not an error: the hover list of further crossings shows it.

### 4.3 Lapse rates, thickness and wind

**Lapse rates** Temperature fall per km between the surface and 3 km, 850–500 and 700–500 hPa. The ISA has 6.5 K/km; values from 7 (amber) towards the dry adiabatic 9.8 K/km (red from 9) mean steep, potentially unstable layers. Small or negative values mean stable air or inversions.

**1000–500 hPa thickness** The depth of the layer between the two pressure surfaces, in decametres. By the hypsometric equation it depends only on the layer's mean temperature: 1 dam corresponds to about 0.5 K. 551 dam means a layer mean of about −1.5 °C; about 528 dam is a common rule-of-thumb divide between rain and snow. Thickness is an excellent comparison between the aircraft and the sonde, because it averages the whole lower half of the troposphere: on 14–19 October the two agreed to +0.6 dam on average.

**T 850 / 700 / 500 hPa** Temperatures at the standard pressure levels used on weather maps (sonde: and dewpoints).

**Bulk shear 0–1, 0–3, 0–6 km** The vector difference between the wind at that height above the ground and the lowest layer. Strong 0–6 km shear ($\geq 40$ kt, amber) organises convection when there is any.

**Max wind, low-level jet** The strongest wind (a jet stream from 60 kt) and a wind maximum below 5 000 ft that is at least 25 kt and 10 kt stronger than above it.

**Lowest inversion, turbulence layers** From the automatic analysis: the lowest significant inversion with its temperature rise, and layers with a turbulence risk from strong shear and a small Richardson number.

**Advection** The strongest warm or cold advection, estimated from the turning of the wind with height (the thermal wind, Section 5.4).

The area profile is a 150 km, 60-minute average, so its shear and maximum wind are usually smaller than in the sonde: on 14–19 October the Mode S 0–6 km shear was on average 8 kt lower.

# 5 The time–height section

A single sounding is a snapshot. The weather is a film: air masses, fronts, troughs and ridges move over the receiver, and the profile changes from hour to hour. The Time–height view shows this film on one picture. It is built from the hourly profile archive: **one column per UTC hour, one row per altitude layer, time running to the right.** The current hour is added live from the observations (dashed frame).

## 5.1 How to read the picture

- **Height scale.** The vertical axis is the square root of the pressure altitude, which stretches the lowest levels: the 1 000 ft layers below 5 000 ft keep a readable height, while FL300–FL420 are compressed. The left axis gives feet and flight levels, the right axis the ISA pressure.
- **Colour field** (selector): temperature anomaly, temperature change over 3 h, wind speed, lapse rate, vertical motion, vorticity, height of the pressure surfaces, height change over 6 h or the thickness-temperature check (below).
- **Barbs**: wind barbs on every second column and layer as space allows, with a dark halo so they read on any colour.
- **Isotherms**: the lowest 0 °C crossing of each hour (solid) and the −10 and −20 °C crossings (dashed, dotted), joined from hour to hour.
- **Markers**: pink triangles at the top mark the Jokioinen launches; day separators carry the weekday.
- **Aircraft strip**: below the plot, the number of aircraft in each hour; red bars ($< 8$ aircraft, LOW) and amber bars ($< 20$) warn that the column rests on little data — typically 22–02 UTC. Cells with fewer than 3 replies are drawn faint; empty cells (dark) had no data at all.
- **Hover and click**: the tooltip gives the hour, the layer and its pressure, temperature with anomaly and 3-hour change, lapse rate, wind and aircraft counts (Fig. 5). A click opens that hour on the Skew-T (Area · hour), with its indices and the sonde.
- **Length and end**: 24 h, 48 h, 72 h, 7 or 14 days, ending now or at the end of a chosen UTC day.

**Time is (almost) space.** Weather systems move, mostly with the wind of the middle troposphere. What passes over the receiver during a day was upstream the day before, so a time–height section is close to a vertical cross-section through the weather system along its direction of motion — with one twist: because time runs to the right, the part of the system that arrives *first* (its leading, downstream side) is on the *left*. A system moving at 20 kt covers about 900 km in 24 hours; the 300 km wide averaging area corresponds to several hours (about 8 hours at 20 kt), so sharp features are smoothed over several columns.

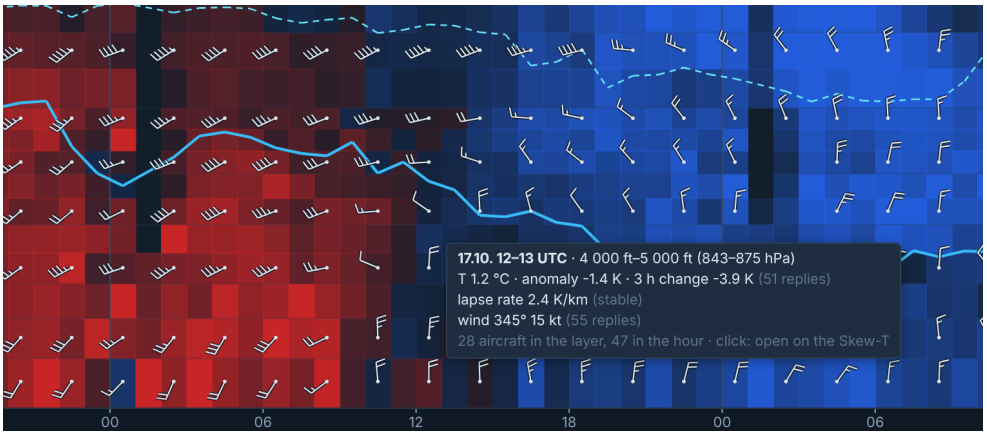



**Figure 5.** Tooltip on 17 October 12–13 UTC at 4 000–5 000 ft, just behind the cold front: 1.2 °C, 1.4 K colder than the period mean and 3.9 K colder than three hours earlier, stable (2.4 K/km), wind 345°/15 kt, 28 aircraft in the layer.

## 5.2 Temperature anomaly: air masses and fronts

The **temperature anomaly** is each layer's temperature minus the mean of the same layer over the shown period (blue colder, red warmer, full colour at ±6 K). Subtracting the layer mean removes the normal fall of temperature with height, so what remains are the air masses. Note that the reference is the *shown period*: in a 72-hour view a front is a strong contrast, in a 14-day view the same hours may be only part of a longer warm or cold spell. Change the length to change the reference.

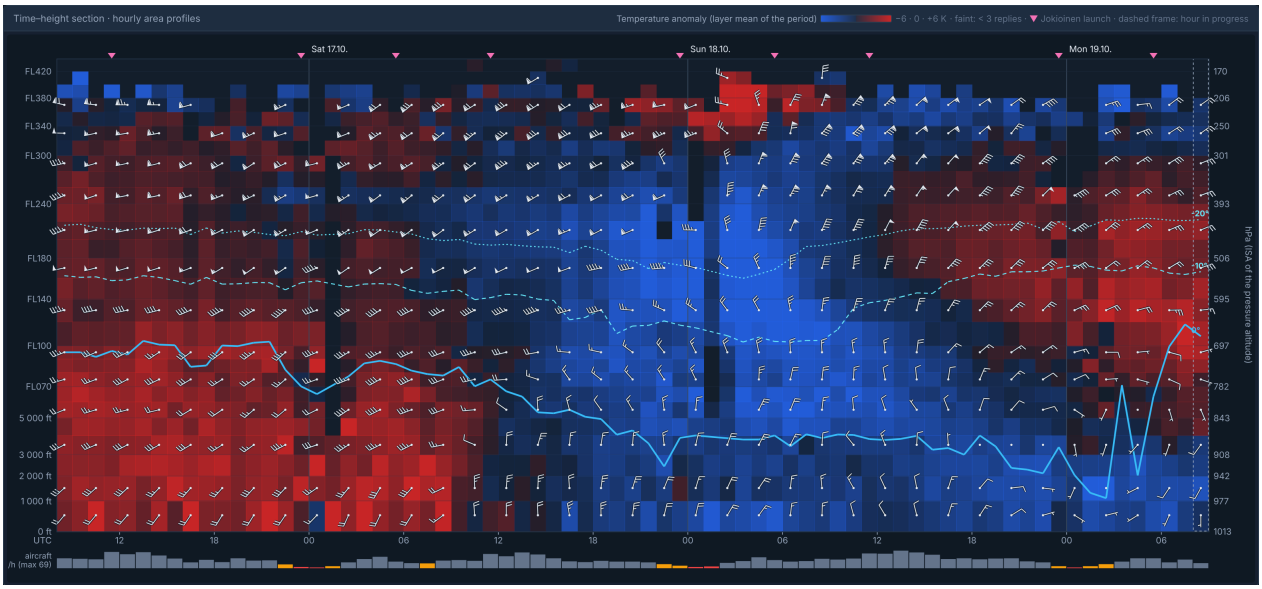



**Figure 6.** Temperature anomaly, 72 hours to 19 October 2026 08 UTC, with barbs and isotherms. Warm air (red) with south-westerly winds on 16–17 October; the cold front arrives near the ground at about 10 UTC on 17 October and higher up later, so the boundary rises to the right; the 0 °C line falls from about FL100 to 3 500 ft. Early on 18 October a lowered tropopause makes FL340–FL380 warm. From the afternoon of 18 October the middle troposphere warms (sinking air under the high) while the lowest 3 000 ft stay cold.

**Reading the cold front of 17 October** (Fig. 6). The hourly area means show the cold air arriving at different times at different heights:

- 0–4 000 ft: cooling of 4–6 K between 08 and 12 UTC, and the wind turning from south-west (244°/21 kt at 1 000 ft at 08 UTC) to north (358°/23 kt at 10 UTC);
- 6 000–10 000 ft: cooling mainly between 12 and 18 UTC, the wind turning from south-west to west and north-west between about 14 and 18 UTC;
- FL140–FL240: cooling between 14 and 22 UTC.

The boundary therefore climbs about 20 000 ft in 10–12 hours. This is the wedge of Fig. 2 seen in time: a cold front's cold air is shallow at its leading edge and deepens behind it. If the front moved at about 20 kt, 11 hours correspond to about 400 km, a slope of roughly 1:70 — within the 1:50 to 1:100 typical of cold fronts. A *warm front* would look the other way round: the warm air arrives aloft first and reaches the ground hours later, so its boundary *descends* to the right.

**Troughs and the tropopause.** Behind the front, early on 18 October, the layers at FL340–FL380 turn red: at FL380 the temperature rose from about −58 to −50 °C. In the cold air behind a trough the tropopause is low, and the air at these levels is now stratospheric — warmer than the tropospheric air that is normally there. A warm patch at the top of the section above cold air below is the signature of a lowered tropopause.

**Sinking air under a high.** From the afternoon of 18 October the QNH rose from 1031 to 1040 hPa and the middle troposphere warmed strongly: at 10 000 ft from −11 °C (08 UTC on the 18th) to −1 °C (06 UTC on the 19th), at 6 000 ft by 6–7 K, while the lowest 2 000 ft hardly changed. Air sinking in an anticyclone is compressed and warms at the dry adiabatic rate; descent of a few hundred metres to a couple of kilometres a day is typical. The warming reaches down only to the top of the cool, moist boundary layer, so a *subsidence inversion* forms between them (Section 5.5).

**The daily cycle.** In the lowest 1 000–3 000 ft the anomaly alternates between day and night on clear days: daytime heating mixes up to the dry thermal top, nights cool the lowest layers.

### 5.3 Temperature change over 3 hours: advection as it happens

The **3-hour change** is each layer's temperature minus that of the same layer three hours earlier (full colour at ±4 K). It does not depend on the period, and it shows *when* things happen. The local change of temperature has three causes:

$$\frac{\partial T}{\partial t} = \underbrace{-\vec{V}\cdot\nabla T}_{\text{advection}} + \underbrace{(\Gamma_d - \Gamma)\,w}_{\text{vertical motion}} + \underbrace{Q}_{\text{heating, cooling}}$$

*Advection* is warmer or colder air carried in by the wind; *vertical motion* warms sinking air and cools rising air (in proportion to how much less steep the actual lapse rate $\Gamma$ is than the dry adiabatic $\Gamma_d$); *diabatic* heating and cooling are the sun, night-time radiation, condensation and evaporation. Near the ground the daily cycle dominates; above about 5 000 ft advection and vertical motion do.

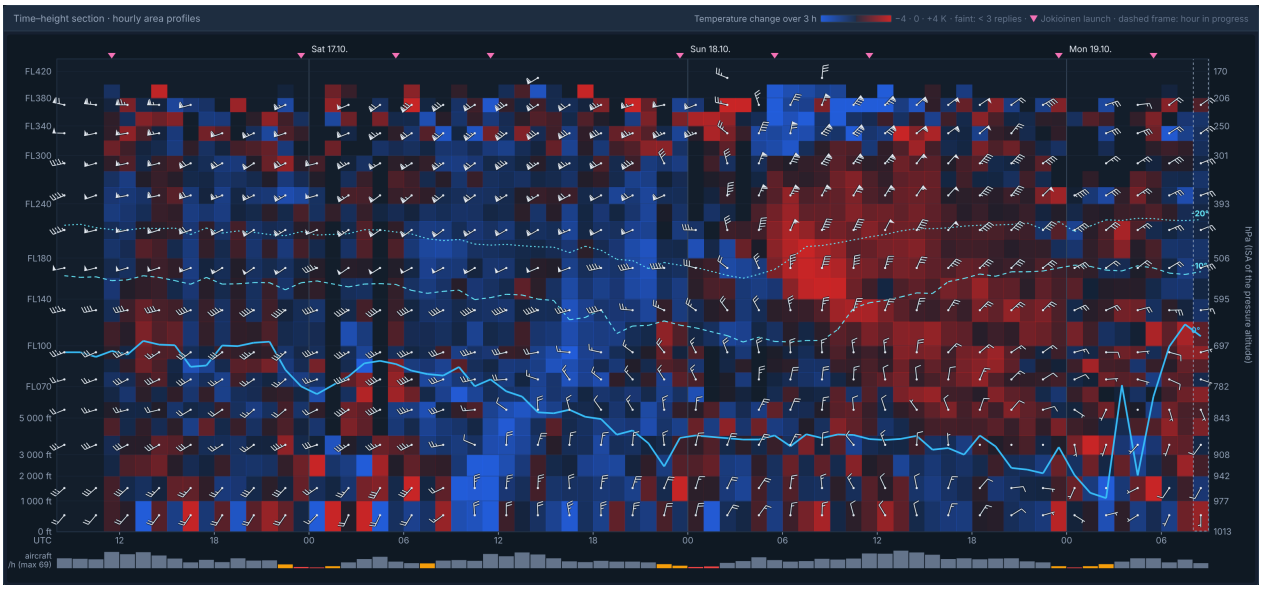



**Figure 7.** Temperature change over 3 hours, same period. The cold front is a blue band rising from the ground at about 10 UTC on 17 October to FL240 around midnight; the sinking air of 18 October is a broad red area at FL100–FL240 from about 05 to 15 UTC. Near the ground, alternating red and blue cells follow the day and night.

The change field is noisier than the anomaly, because a single hour's layer mean scatters by about 1 K and the change subtracts two of them. Look for coherent areas several hours and layers wide.

### 5.4 Wind speed: jets and the thermal wind

The **wind speed** field colours each cell in 10 kt steps, from dark blue (calm) through green and yellow to red and purple (130 kt and more). It shows the jet streams — the strongest winds, usually just below the tropopause — their height and their timing, low-level jets near the ground and calm layers.

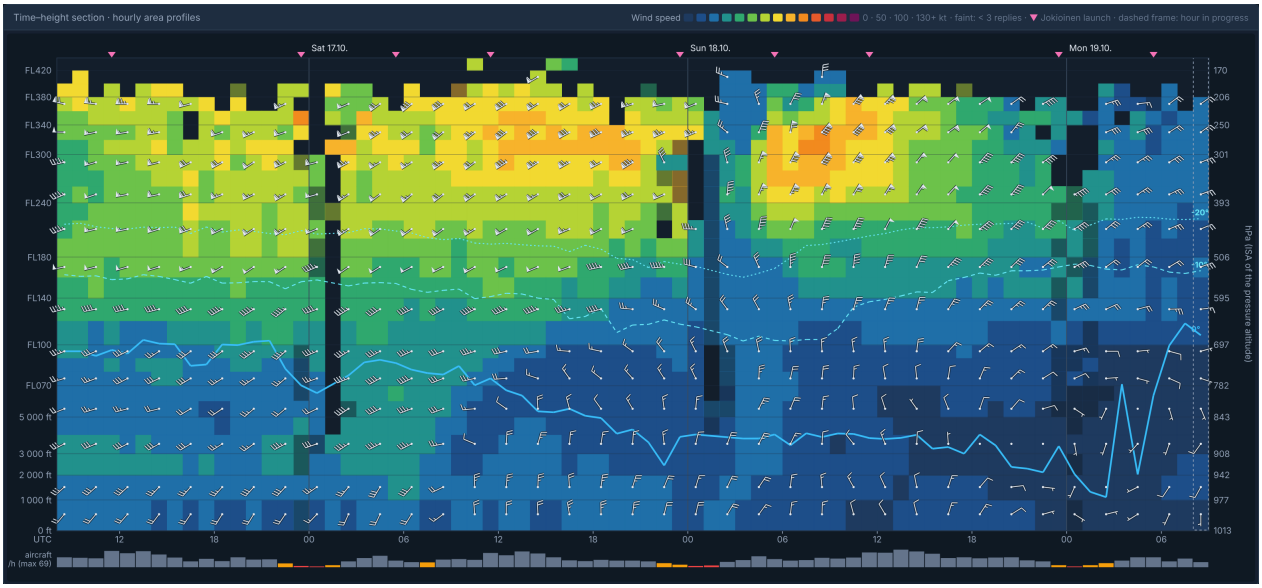



**Figure 8.** Wind speed, same period. A south-westerly jet of 70–95 kt at FL300–FL340 lies above the warm air and the front on 16–17 October, and a 30–35 kt south-westerly flow at 1 000–4 000 ft ahead of the front. On 18 October a north-easterly jet of up to 95 kt passes at FL300 (07–08 UTC). Under the high the winds become light at all levels.

**The thermal wind.** The barbs carry more than speed. Because the wind in the free atmosphere is close to geostrophic — it blows along the height contours of the pressure surfaces — its change with height is tied to the horizontal temperature gradient (the *thermal wind*): the vertical wind shear blows along the isotherms of the layer, with cold air to the left. Two practical rules follow:

- the strongest winds lie above the strongest temperature contrasts — the jet of 17 October sits above the front;

- a wind that **veers** (turns clockwise) with height means warm advection; one that **backs** (turns counter-clockwise) means cold advection. At 12 UTC on 17 October the wind backed from 004° at 1 000 ft through 345° at 4 000 ft to 257° at 8 000 ft: cold advection, exactly what the change field shows.

The Skew-T Analysis (level Advection) and the Advection row of the Indices panel turn this rule into numbers in K/h.

### 5.5 Lapse rate: stability and inversions

The **lapse rate** field shows how fast temperature falls with height at each layer, from the temperatures of the layers below and above, smoothed over ±1 hour (a single hour's layer means scatter by about 1 K, which over 1 000 ft would be 3 K/km of noise). The colours are classes in K/km: violet inversion ($< -1$), dark blue very stable ($-1$ to 2), light blue stable (2–5), grey normal (5–7), orange steep (7–9) and red dry-unstable ($\geq 9$, the dry adiabatic rate is 9.8).

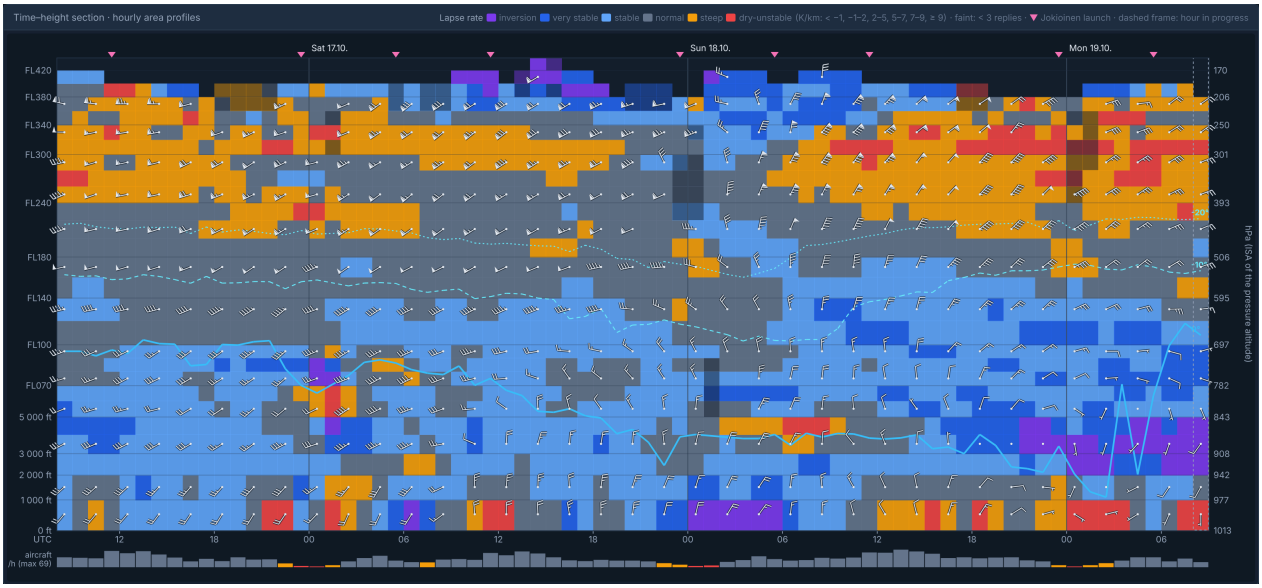



**Figure 9.** Lapse rate, same period. A night-time surface inversion (violet at 0–1 000 ft) in the early hours of 18 October; steep, near-adiabatic lapse rates near the ground on the sunny afternoon of 18 October; the subsidence inversion forming at 3 000–5 000 ft from the evening of 18 October; steep upper-tropospheric lapse rates below the tropopause and a very stable band at about FL380 (the tropopause). The red cells near the ground around midnight of 19 October rest on 2–8 aircraft (red bars in the strip) and are noise.

Stability decides much of what the weather does:

- **Inversions** stop vertical mixing. Below them haze, fog and low stratus are trapped; a sharp change of wind across the inversion top gives low-level wind shear. Night-time *radiation inversions* form from the ground up on clear, calm nights and are mixed away by the morning sun. *Subsidence inversions* form aloft under a high, as on 18–19 October, and can last for days. *Frontal inversions* separate the cold air under a front from the warm air above (the warm nose at 1 500 ft in Fig. 2).
- **Steep lapse rates** mean that rising air stays warmer than its surroundings: thermals near the ground, convection and showers when the air is moist, turbulence.
- The **tropopause** appears as the change from orange or grey to blue at the top of the section: the stratosphere above is very stable.

### 5.6 Vertical motion: rising and sinking air

Weather is made by vertical motion: rising air cools, forms cloud and precipitation; sinking air warms and dries, and clears the sky. Large-scale vertical motion is slow — typically 1–5 cm/s, a few hundred metres to a few kilometres a day — and cannot be measured directly by aircraft or radiosondes. It can be inferred from its effect on temperature.

**The adiabatic method.** Air that rises expands and cools at the dry adiabatic rate $\Gamma_d$ = 9.8 K/km; the surrounding air cools with height only at the actual lapse rate $\Gamma$. A parcel lifted by $\Delta z$ therefore makes the layer colder by $(\Gamma_d - \Gamma)\,\Delta z$, and sinking air warms it by the same amount. This is the vertical-motion term of the temperature equation of Section 5.3. Leaving out heating and cooling and solving for $w$:

$$w = \frac{-\vec{V}\cdot\nabla T - \partial T/\partial t}{\Gamma_d - \Gamma}$$

— the part of the local temperature change that the wind does not bring in must be ascent or descent. All three ingredients

are in the hourly archive:

- the **advection** $-\vec{V}\cdot\nabla T$, from the turning of the wind between the layers two below and two above (the thermal wind, Section 5.4);
- the **local change** $\partial T/\partial t$, from the temperatures 3 hours before and after;
- the **lapse rate** $\Gamma$, from the layers next to it.

Temperatures and winds are smoothed over $\pm 1$ hour first. The field is drawn between 5 000 ft and FL200 in green (rising) and orange (sinking), full colour at $\pm 3$ cm/s. The tooltip shows the value with its two ingredients in K/h.

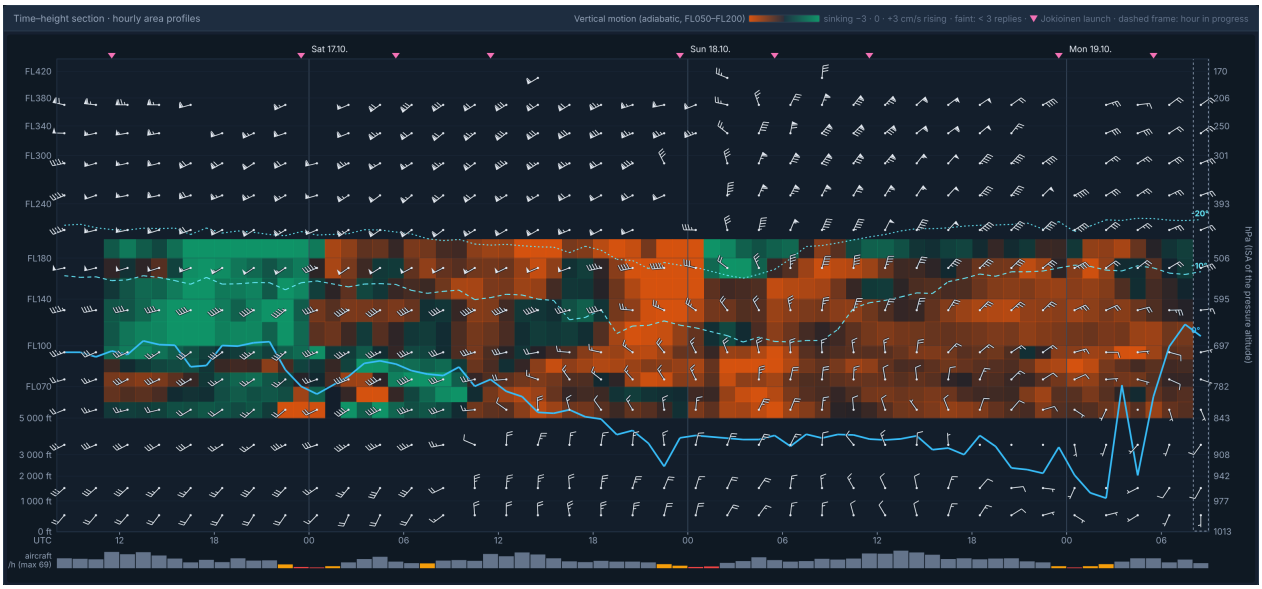



**Figure 10.** Vertical motion, 72 hours to 19 October 2026 08 UTC. Rising air (green, up to about +5 cm/s) in the warm south-westerly air ahead of the front on 16 October; sinking behind the front from 17 October; a short burst of ascent at FL160–FL200 early on 18 October as the trough aloft passes; steady sinking of about 0.5–3 cm/s under the building high on 18–19 October — the subsidence that warmed the middle troposphere in Fig. 6. The first 3 hours have no value (no earlier data in the shown period); the newest hours are fainter.

**How good is it?** Tested on 24 September – 20 October 2026 between 5 500 ft and FL190: the values are typically 2–3 cm/s, the normal size of synoptic vertical motion. Computing $w$ separately from two random halves of the aircraft gave the same picture (correlation 0.65–0.83), with a noise of about 0.6–1 cm/s. It agrees with the weather: rising air in 59 % of the hours with broken cloud but only 30 % of the clear ones, and a correlation of $-0.47$ with the 6-hour change of the QNH — falling pressure, rising air.

**Limits.** The method assumes no heating or cooling. Below about 5 000 ft the sun and night-time radiation dominate, so nothing is drawn there. In cloud, condensation releases heat and the ascent is underestimated. Above FL200 $\Gamma$ approaches $\Gamma_d$ and the wind shear grows, so small errors become large values; nothing is drawn there either, nor where $\Gamma_d - \Gamma < 1.5$ K/km. The newest hours have no "3 hours after" yet; they use the latest available hour instead and are drawn fainter, and the hour in progress has none.

### 5.7 Vorticity: the spin of the wind field

**Relative vorticity** $\zeta = \partial v/\partial x - \partial u/\partial y$ measures how much the wind field rotates: positive (anticlockwise in the northern hemisphere) is *cyclonic*, negative *anticyclonic*. It comes from curved flow — troughs and lows are cyclonic, ridges and highs anticyclonic — and from shear across the flow: the cold side of a jet stream is cyclonic, the warm side anticyclonic. The Earth's own spin adds the Coriolis parameter $f = 2\Omega\sin\varphi = 12.7\times10^{-5}\ \mathrm{s}^{-1}$ at 60°N; synoptic relative vorticity is typically a few to $15\times10^{-5}\ \mathrm{s}^{-1}$, and values comparable to $f$ mark a sharp trough.

Vorticity matters because it travels with the air and changes in a predictable way. Where the upper wind carries the cyclonic vorticity of a trough towards the ridge ahead of it (*positive vorticity advection*), the air aloft diverges and the air beneath it rises; behind the trough the opposite happens and the air sinks — which is why cloud and precipitation lie ahead of upper troughs and clearing skies behind them. Shown next to the vertical-motion field, it lets you see both halves of that relation.

**How it is computed.** Every hour, for each altitude layer, the winds of all aircraft within 200 km of the receiver are averaged per aircraft and fitted with a plane $u, v = a + bx + cy$; the slopes give $\zeta = \partial v/\partial x - \partial u/\partial y$ and its standard error. A layer needs at least 6 aircraft spread across the area in two directions — below about FL130 the aircraft follow

the airport's approach and departure paths and rarely qualify, so the field normally covers about FL130–FL390 (lower in busy hours). Aircraft whose wind departs from the plane by more than 3 times the RMS are left out. The values are stored for every hour (also after the observations are purged), and the hour in progress is computed live. The colours are purple (cyclonic) and gold (anticyclonic), full at $\pm 15 \times 10^{-5}\ \mathrm{s}^{-1}$; cells with an error above 4 or fewer than 10 aircraft are faint.

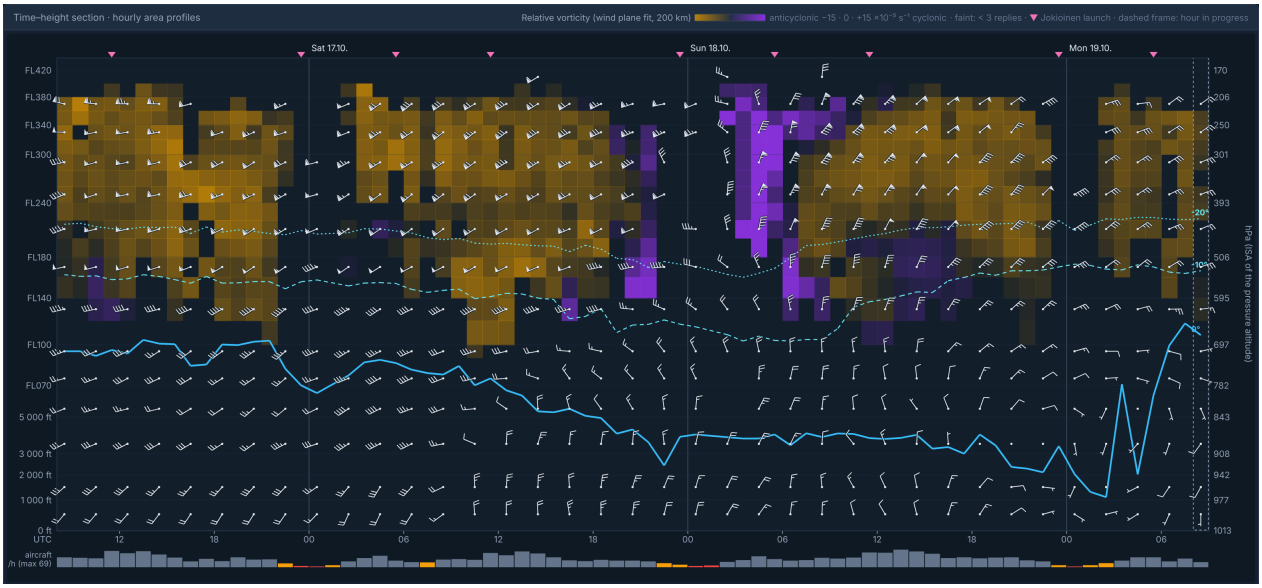



**Figure 11.** Vorticity, same period. Anticyclonic flow (gold, about −5 to −11) in the ridge ahead of the front and under its south-westerly jet on 16–17 October; two cyclonic columns behind the front — in the evening of 17 October (up to +14 at FL140) and at 03–05 UTC on 18 October (+20 to +40 at FL200–FL340; the largest values rest on only 8–10 night-time aircraft, but at 04 UTC +28 at FL200 came from 28 aircraft), the passage of the upper trough, at the same time as the lowered tropopause of Fig. 6; anticyclonic again under the high.

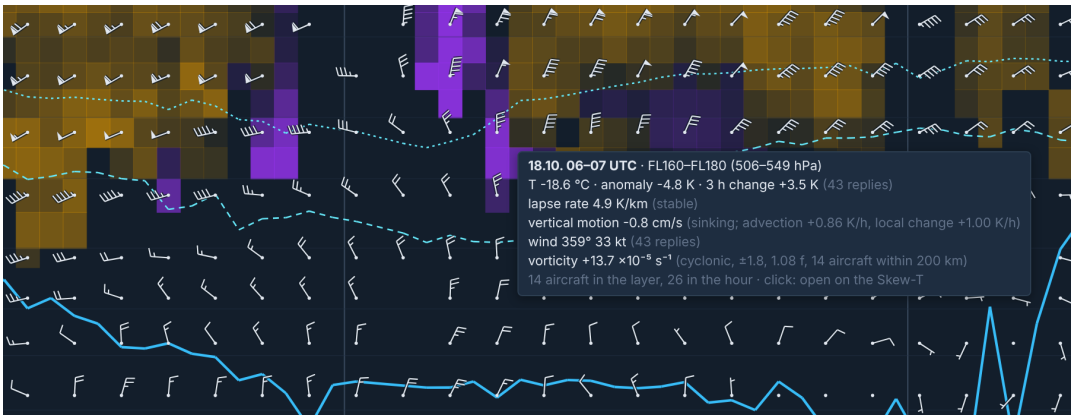



**Figure 12.** Tooltip at 06–07 UTC on 18 October, FL160–FL180, behind the trough: vorticity $+13.7 \times 10^{-5}\ \mathrm{s}^{-1}$ ($\pm 1.8$, about 1.1 $f$, 14 aircraft); the vertical-motion line shows that the strong warming of +1.0 K/h is almost all warm advection (+0.86 K/h), leaving weak sinking of −0.8 cm/s.

**How good is it?** Tested on 24 September – 20 October 2026 (200 km, one hour): the vorticity varies with an RMS of $7–10 \times 10^{-5}\ \mathrm{s}^{-1}$ against an error of about 2.3, and consecutive hours correlate by 0.65–0.83 — a real, slowly changing field. The *divergence* from the same fit (the slope that gives vertical motion by mass balance) is computed and stored but not shown: its hour-to-hour correlation was only 0.1–0.25, i.e. mostly noise.

**One caveat.** Averaged over the whole period, the vorticity came out anticyclonic at all levels, by about 2–4 units. Part of that may be weather (several highs), but climbing and descending aircraft alone gave clearly different averages, which the atmosphere cannot do — so 2–3 units are probably an effect of how the departure and arrival corridors sample the circle. A common heading error was ruled out against the radiosonde: across 49 000 comparisons the aircraft winds differed from the sonde by only +0.7 kt across the track. Read the field for its changes and its strong cyclonic and anticyclonic areas, and treat values within about ±3 of zero as uncertain.

### 5.8 Heights of the pressure surfaces and thickness temperature

**The GNSS height of a pressure surface.** The altitude an aircraft reports is its pressure altitude (Section 2): it tells *which* pressure surface the aircraft is on. Its ADS-B velocity messages, sent about twice a second, also carry the difference between its *GNSS* (satellite) height and that pressure altitude, in 25 ft steps. Pressure altitude plus this difference is the true height of the pressure surface at the aircraft. How far that lies above or below the surface's height in the standard atmosphere is the **D-value** — the quantity of an upper-air height chart: warm air columns and highs lift the pressure surfaces (D positive), cold air and lows lower them. The heights of the surfaces and their slope are what drive the wind: the geostrophic wind blows along the height contours, faster where they are closer together (Section 7.6).

**Calibration against the radiosonde.** The raw difference is not the D-value yet. The aircraft's GNSS height is measured from the ellipsoid, about 18 m above mean sea level in southern Finland (the geoid), and the aircraft's own pressure measurement has a small error that grows with speed. Against the Jokioinen sondes of 20–21 October 2026 the raw aircraft heights were about 24 m too high near the ground, 0 m at FL250–FL330 and 6 m too low above; each airframe has its own offset of typically ±9 m, of which only about 40 % repeated from one day to the next. The correction is therefore learnt continuously: after every Jokioinen sounding, the aircraft within 150 km that passed a level within one hour of the balloon are compared with the sounding's measured height of that pressure surface, and a correction per 5 000 ft band (plus a small per-airframe offset) is fitted again from the last 30 days. It starts from built-in values and follows the data as they accumulate. Only aircraft with good GNSS (NACp $\geq 8$) outside their GPS-degradation episodes are used — jammed aircraft usually stop sending the difference anyway, so the night hours, when jamming is common, have few values.

**Height of the pressure surfaces.** Each cell shows the D-value of the layer: the median over the airframes of the hour, red where the surface lies higher and blue where it lies lower than in the standard atmosphere, full colour at ±300 m (Fig. 13). The tooltip gives the height of the surface above mean sea level, the D-value, its 6-hour change and the spread between airframes. Under the high of 19–21 October the D-value was about +120 to +200 m near the ground but +230 to +420 m at FL240–FL300: the column was warm, so the upper surfaces were lifted more than the lower ones; all of it fell with time as the high weakened.

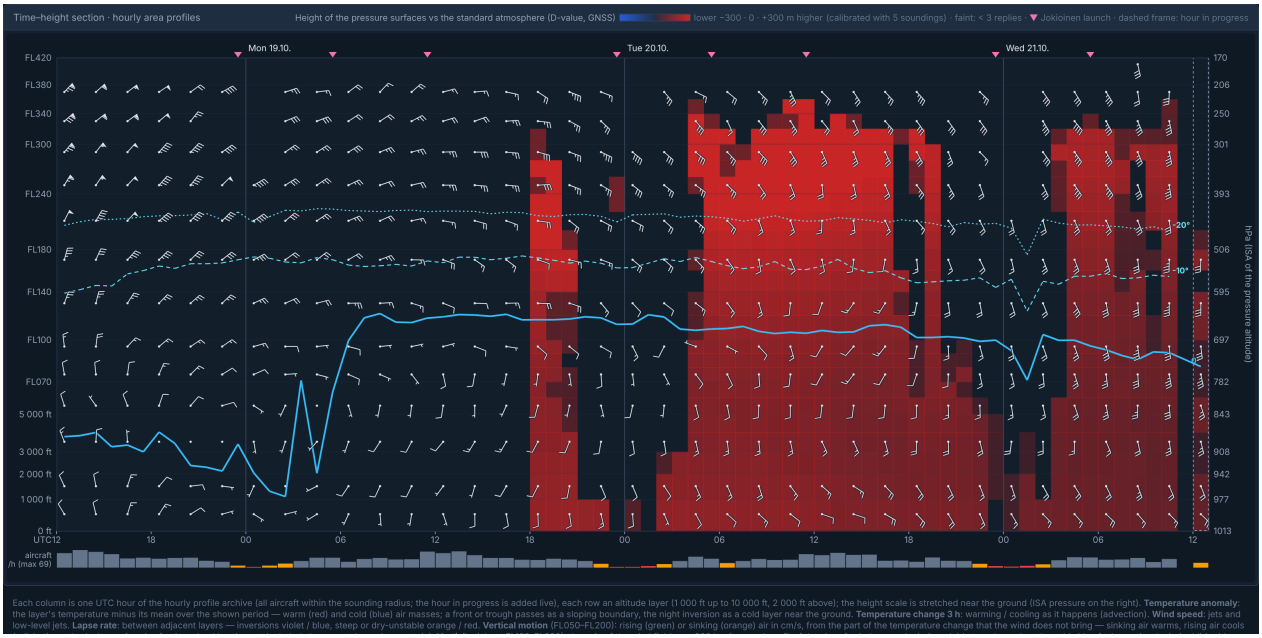



**Figure 13.** Height of the pressure surfaces, 72 hours to 21 October 2026 12 UTC (the GNSS heights are recorded from 19 October 18 UTC). The whole column lies higher than standard under the high (QNH 1029–1040): D about +120–+200 m near the ground and +230 to +420 m above FL240, lower towards 21 October. The empty night columns are hours with GNSS jamming.

**Height change 6 h.** The D-value minus that of the same layer six hours earlier (full colour at ±60 m): surfaces falling ahead of a trough or a front (blue), rising behind it and in a building high (red). It is the time–height version of the pressure tendency at the ground, at every level (Fig. 14).

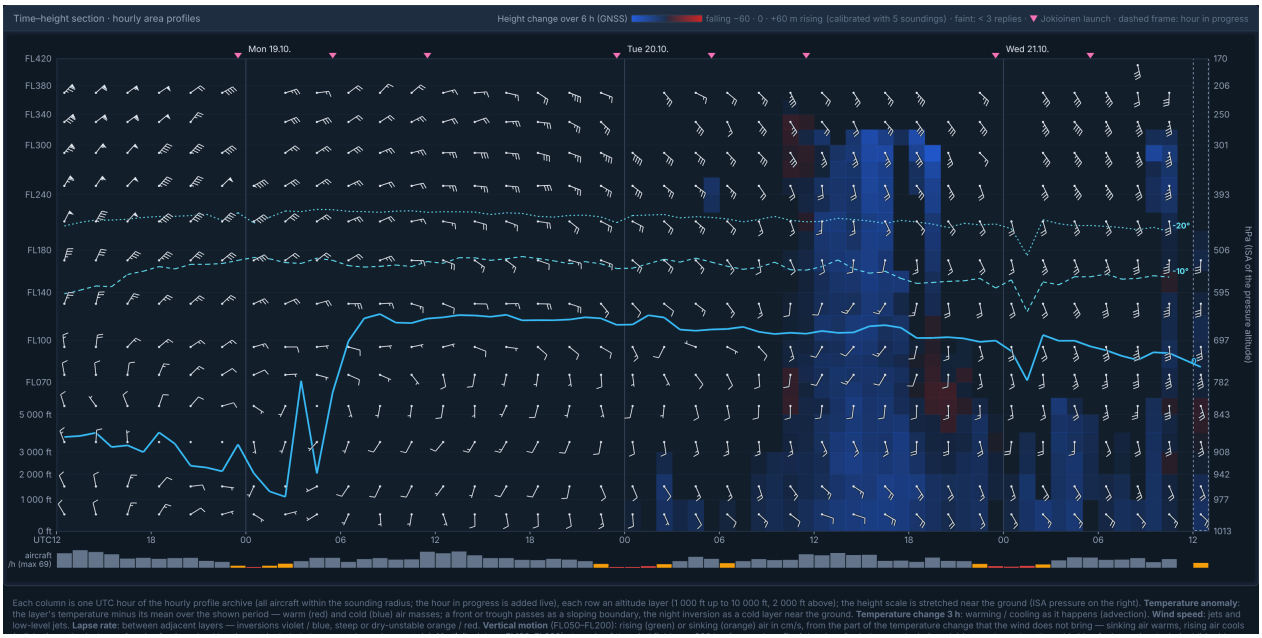



**Figure 14.** Height change over 6 hours, same period. On 20 October the surfaces fall by 20–40 m per 6 hours at all levels as the high weakens; at the ground the METAR QNH fell from 1037 to 1032 hPa over the same hours, about 40 m.

**Thickness temperature check.** A climbing or descending aircraft measures the true height of the pressure surface at the bottom and at the top of a layer within minutes. By the hypsometric equation the height difference depends only on the mean (virtual) temperature of the layer:

$$\bar{T}_v = \frac{g_0 \, \Delta H}{R \ln(p_1/p_2)},$$

with $\Delta H$ the difference of geopotential heights. This temperature is measured in a completely different way from the Mach/TAS temperature, and the airframe's own offset cancels between the two ends. For each 5 000 ft layer the field shows the thickness temperature of the hour's climbs and descents *minus* the Mach/TAS temperature of the same layers (full colour at ±2 K; Fig. 15). Near 0 means the two methods agree. Moist air near the ground makes the thickness temperature a few tenths of a kelvin warmer (it is a *virtual* temperature). One flight gives the layer mean to about ±0.8 K for 5 000 ft and ±0.5 K for FL100–FL300; the tooltip (Fig. 16) also gives the bulk FL100–FL300 value.

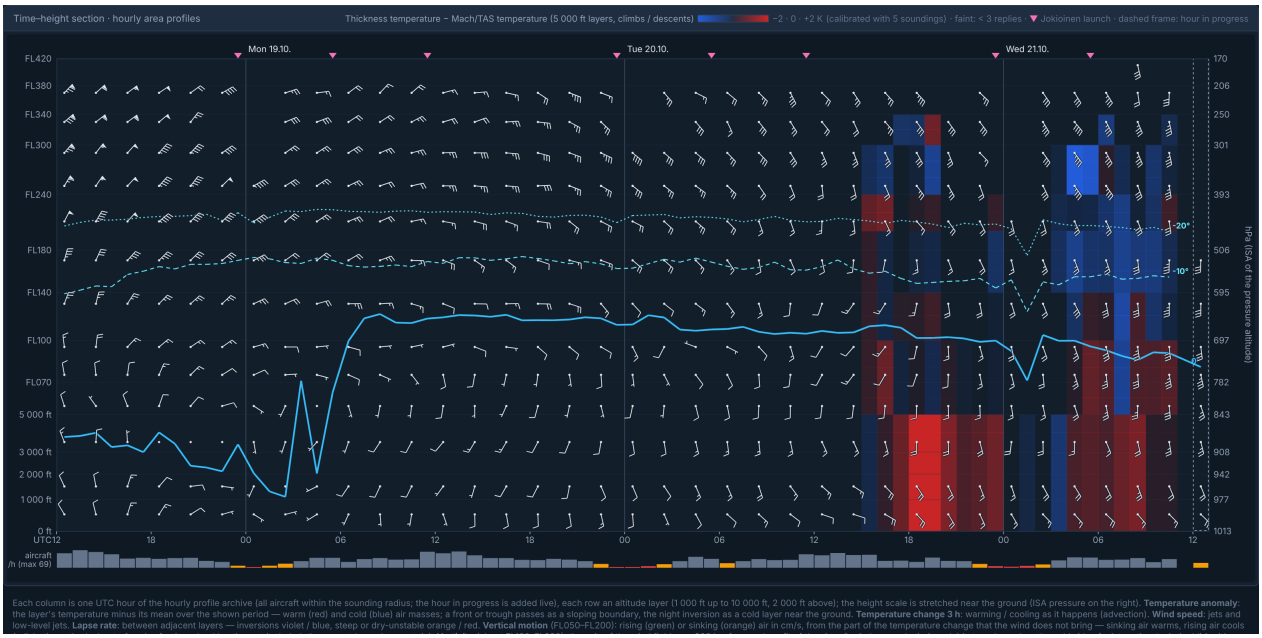



**Figure 15.** Thickness temperature check, same period (the dense climb and descent data are recorded from 20 October 11 UTC). Most cells are within ±1 K; the warmer cells in the lowest 5 000 ft on the evening of 20 October and the colder ones at FL250–FL300 early on 21 October are among the things the calibration will be checked on as data accumulate.

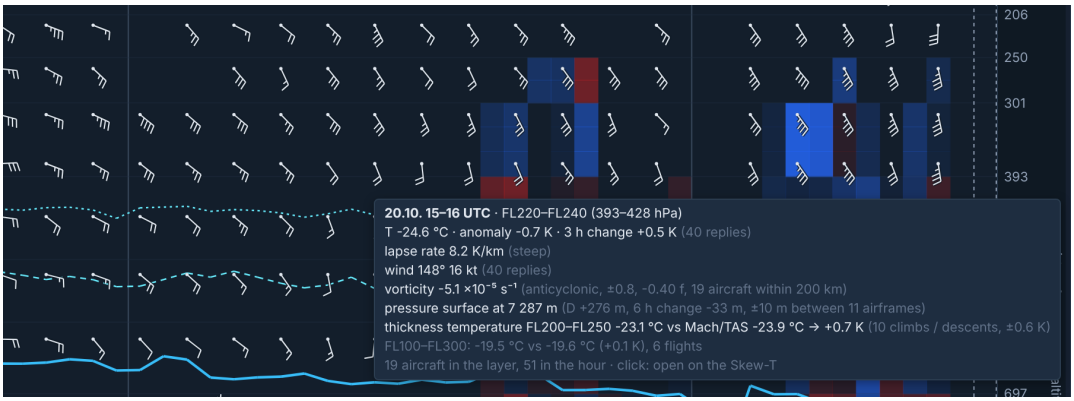



**Figure 16.** Tooltip at 15–16 UTC on 20 October, FL220–FL240: the pressure surface at 7 287 m (D +276 m, 33 m lower than 6 hours earlier, ±10 m between 11 airframes); the 5 000 ft layer FL200–FL250 from 10 climbs and descents −23.1 °C against −23.9 °C from Mach/TAS; FL100–FL300 −19.5 against −19.6 °C from 6 flights.

**Status.** These fields are new (October 2026) and rest on a few days of data and soundings so far; the calibration improves by itself with every sounding. Read the height change and the large-scale pattern of the D-value with confidence; treat single thickness cells beyond about ±1 K with care.

### 5.9 Moisture (trial)

**A first trial.** This field was added on 21 October 2026 and rests on one and a half days of data and a handful of soundings. It is an experiment: the method will be tuned as soundings in varied weather accumulate, and its readings should be taken as a hint, not a measurement.

**Why it is possible at all.** Aircraft report no humidity: the Mode S register that can carry it (BDS 4,4) is answered by very few aircraft, and MODE-S Wind has never received it. But moist air is *lighter* than dry air at the same temperature and pressure (a water molecule weighs less than the nitrogen and oxygen it replaces), so a layer of moist air is a little *thicker* than a dry one. The thickness temperature of the previous section is the layer's mean *virtual* temperature $T_v$ — the temperature dry air would need to be as light — while the Mach/TAS temperature is the real temperature $T$. Their difference is the moisture of the layer:

$$T_v - T \;\approx\; 0.608\, q\, T,$$

with $q$ the specific humidity (kg of water vapour per kg of air). The same principle was used with ADS-B heights at Amsterdam Schiphol (EGUsphere preprint 2026-717); here the heights come from the calibrated GNSS heights of Section 5.8.

**How large the signal is.** Very small, and only in the lower troposphere, where the air can hold enough water. From the 34 Jokioinen soundings of 11–21 October 2026:

| Layer | $T_v - T$, humid (RH $\geq$ 70 %) | $T_v - T$, dry (RH $<$ 40 %) | difference |
|---|---|---|---|
| 1 000–5 000 ft | 0.86 K | 0.24 K | 0.6 K |
| 5 000–10 000 ft | 0.64 K | 0.20 K | 0.45 K |
| 10 000–15 000 ft | 0.45 K | 0.14 K | 0.3 K |
| above FL150 | $<$ 0.3 K | $<$ 0.1 K | too small |

In the soundings $T_v - T$ follows the relative humidity closely (correlation 0.84–0.89), so the physics supports an indicator — but only for the three layers in the table. Contrail and ice-supersaturation layers at cruise levels cannot be found this way.

**How the data are gathered.** Every climb and descent within 150 km of the receiver — in practice the traffic of Helsinki-Vantaa and Tallinn — is stored as a dense height series (Section 5.8). From it, for each of the layers 1 000–5 000, 5 000–10 000 and 10 000–15 000 ft:

1. the GNSS height of the pressure surface at the bottom and at the top of the layer: the rows within ±300 ft of the edge (at least 3, one single crossing, NACp $\geq$ 8, outside GPS-degradation episodes), both edges within 25 minutes; corrected with the height calibration;
2. the thickness temperature $T_v$ of the layer from the height difference (the formula of Section 5.8);
3. the Mach/TAS temperature $T$ of the same layer in the same hour (the area profile, weighted by $\ln p$), and the flight's $T_v - T$;

4. whether the aircraft climbed or descended.

The lowest layer starts at 1 000 ft rather than at 0: the 0 ft pressure altitude lies above the runway only when the QNH is high, so in rainy, low-pressure weather the 0–5 000 ft layer would disappear just when it matters.

**How the indicator is calculated.**

1. *Averaging.* One flight scatters by about 1 K at 5 000–15 000 ft and by 1.8 K below 5 000 ft — more than the whole signal. Each hour therefore uses all flights of ±1 hour around it (±2 hours below 5 000 ft) and their 10 % trimmed mean (the highest and lowest tenth left out); at least 6 flights are needed (12 below 5 000 ft).
2. *Offsets from the soundings.* At every Jokioinen sounding of the last 30 days, the flights of ±2 hours around the balloon are compared with the sonde's own $T_v - T$ of the layer, computed from its temperature and dewpoint. The mean difference is the offset of that layer, learnt separately for climbs and descents: below 5 000 ft descents read about 0.6 K higher than climbs, probably because the aircraft's pressure (altimeter) error changes as it slows down and sets flaps. Each offset is shrunk towards zero as if three soundings had shown no difference, so with the first soundings it stays small and then follows the data. Every flight's value is corrected with the offset of its layer and direction before averaging.
3. *Index.* The corrected $T_v - T$ is divided by the value saturated air of the same temperature would give, $0.608\, q_{\mathrm{sat}}(T,p)\, T$:

$$\text{index} \;=\; 100 \times \frac{T_v - T - \text{offset}}{0.608\, q_{\mathrm{sat}}\, T}\ \%,$$

   which is roughly the layer's mean relative humidity. At 5 000–10 000 ft, $0.608\, q_{\mathrm{sat}} T$ is about 0.75 K in October, so an uncertainty of 0.15 K is about ±20 % in the index.
4. *Classes.* **dry** (brown, index $< 40$ %), **moist** (green, 40–70 %), **humid** (blue, $\geq 70$ %); faint where the uncertainty of the index exceeds ±25 %. Values above 100 % or below 0 can occur — they are the noise and remaining offsets of a trial, not supersaturation.

With Sonde on, each sounding gets the same index from its own humidity in its narrow column. For the soundings the index equals the layer's mean relative humidity to within about 1–2 %, so the sonde columns can be read directly as relative humidity.

**Example: the dry high gives way to moist air.** On 18–20 October 2026 the air under the high was very dry (sonde 20 October 12 UTC: 6 % relative humidity at 5 000–10 000 ft); during 21 October moister air arrived on a southerly flow (00 UTC 39 %, 06 UTC 67 %, 12 UTC 59 %). The aircraft followed the change (Fig. 17): at 5 000–10 000 ft brown (dry) on the afternoon of 20 October, green (moist) in the morning of 21 October, blue (humid) from 10 UTC. But they overdid it: at midday the index was 112–125 % against the sonde's 60 % (Figs. 18 and 19), which is a difference of about 0.6 K in $T_v - T$. Whether that is a bias of the method, a real difference between coastal Helsinki and inland Jokioinen 100 km away, or both, needs many more soundings to tell; the offsets have so far been learnt from a single one. At 10 000–15 000 ft the change was weaker, from dry to moist, again more humid than the sonde at midday (index 55–70 % against 34 %). The lowest layer, 1 000–5 000 ft, was noisy: too humid in the evening of 20 October, too dry in the morning of 21 October.

**Figure 17.** Moisture (trial), 48 hours to 21 October 2026 14 UTC, with the soundings as their own columns (pink frames). The dense climb and descent data start on 20 October 11 UTC; nights have too few flights. The sonde columns show the dry air of 20 October and the moistening of 21 October; the aircraft show the same change at 5 000–15 000 ft, but more humid than the sonde at midday on 21 October.

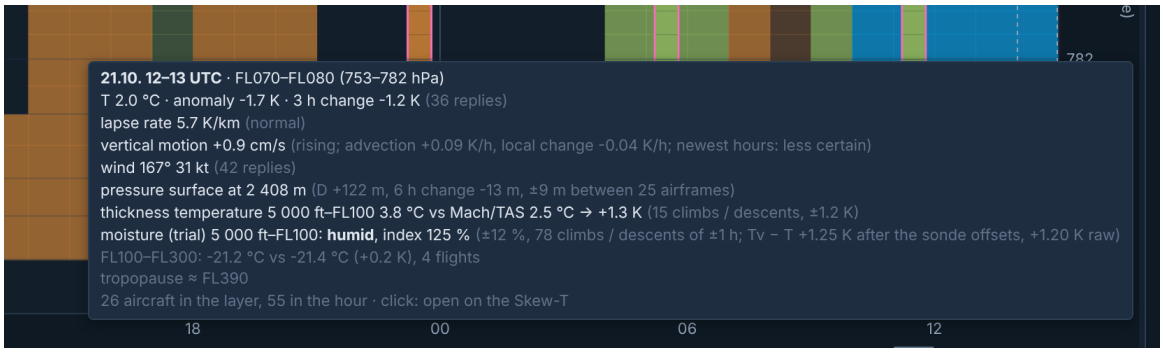



**Figure 18.** Tooltip at 12–13 UTC on 21 October, FL070–FL080: the 5 000–10 000 ft layer is `humid`, index 125 % ±12 % from 78 climbs and descents of ±1 h; $T_v - T$ is +1.20 K raw and +1.25 K after the (still tiny) sonde offsets. The line above it gives the same hour's thickness check from that hour's 15 flights only.

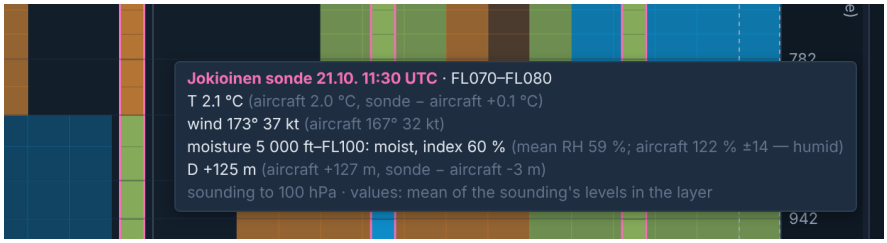



**Figure 19.** Tooltip on the sonde column of 21 October 11:30 UTC: the sounding's index for 5 000–10 000 ft is 60 % (mean relative humidity 59 %, `moist`), the aircraft's 122 % (`humid`).

### How to read it, for now.

- Trust the *changes* more than the values: a layer turning from brown to green or blue as a front or a moist flow arrives, or drying out behind it.
- Compare with the sonde columns whenever a sounding is in the picture; the agreement there is the test of the method.
- Read 5 000–10 000 ft first (largest signal, many flights); 10 000–15 000 ft is weaker; 1 000–5 000 ft is the least reliable so far.

- It covers only where aircraft climb and descend, mostly near Helsinki-Vantaa and Tallinn, and mainly by day. It tells *when* and *at which height* the area is humid — not *where*; it cannot be drawn as a map.

### 5.10 Tropopause, maximum wind, sonde columns and saving the picture

Three overlays and a button complete the section (Fig. 20):

- Tropopause draws the tropopause of each hour as a magenta dashed line (`TP`): the WMO definition — the lowest level above 5 km where the lapse rate falls to 2 K/km or less and the mean lapse rate to every level within the next 2 km stays so — applied to the ±1 h smoothed layer temperatures. The aircraft data usually end at about FL420, so the line appears only where the tropopause is below that and there are data above it; a high tropopause is not found (the Skew-T Analysis and the sonde reach higher).
- Max wind draws the height of the strongest wind above 10 000 ft in each hour, when it is at least 50 kt (yellow, `Vmax`): the level of the jet core and how it rises and falls.
- Sonde draws each Jokioinen sounding as its own narrow column at its launch time, framed in pink and coloured by the same field — temperature anomaly, wind speed, lapse rate or height of the pressure surfaces — after averaging the sounding into the same layers. Hovering it gives the sonde's values against the aircraft column of that hour. Where the sonde column has the same colour as its neighbours, the two systems agree.
- PNG saves the picture with a title line as a PNG file.

The column tooltip also gives the tropopause and the strongest wind of the hour.

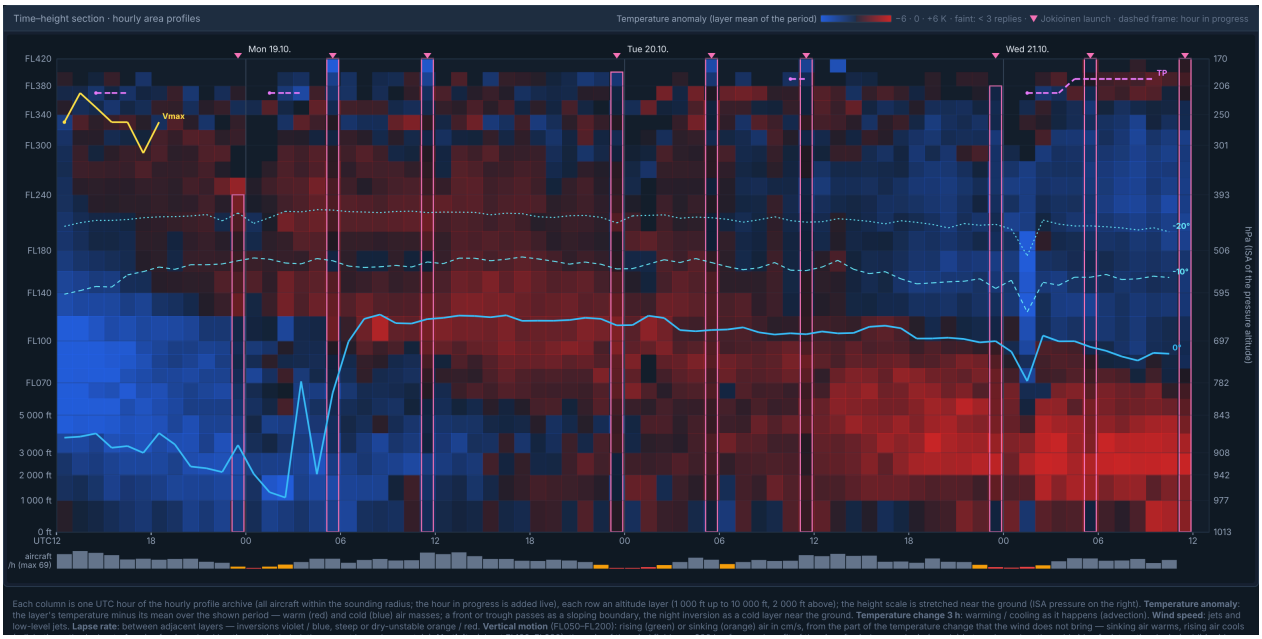



**Figure 20.** Temperature anomaly, 72 hours to 21 October 2026 12 UTC, with Tropopause, Max wind and Sonde (barbs off). The pink-framed columns are the soundings; their colours mostly match the aircraft columns around them. The tropopause is found at about FL370–FL400 in the hours when the data reach above it; winds of 50 kt or more, at FL300–FL380, occur only in the first hours, as the high builds.

### 5.11 Isotherms and the freezing level

The 0, −10 and −20 °C lines are the levels most used in aviation weather: the freezing level for the type of precipitation and for icing, the 0 to −20 °C layer for supercooled water. In Fig. 6 the cold front lowers the 0 °C level from about FL100 to 3 500 ft in 12 hours and the −10 °C level by about 6 000 ft; under the high on 19 October the 0 °C line jumps between 2 000 ft and FL100 because the air is close to 0 °C over the whole lower troposphere.

### 5.12 Longer periods

The 7- and 14-day views show the weather regimes: a series of warm and cold air masses, the jet stream coming and going, and the gradual change of the season (Fig. 21). Barbs are best switched off at this length. The hourly archive keeps every profile, so the section can be drawn for any past period by choosing its end date; hours stored before 11 October 2026 have no aircraft count (no strip).

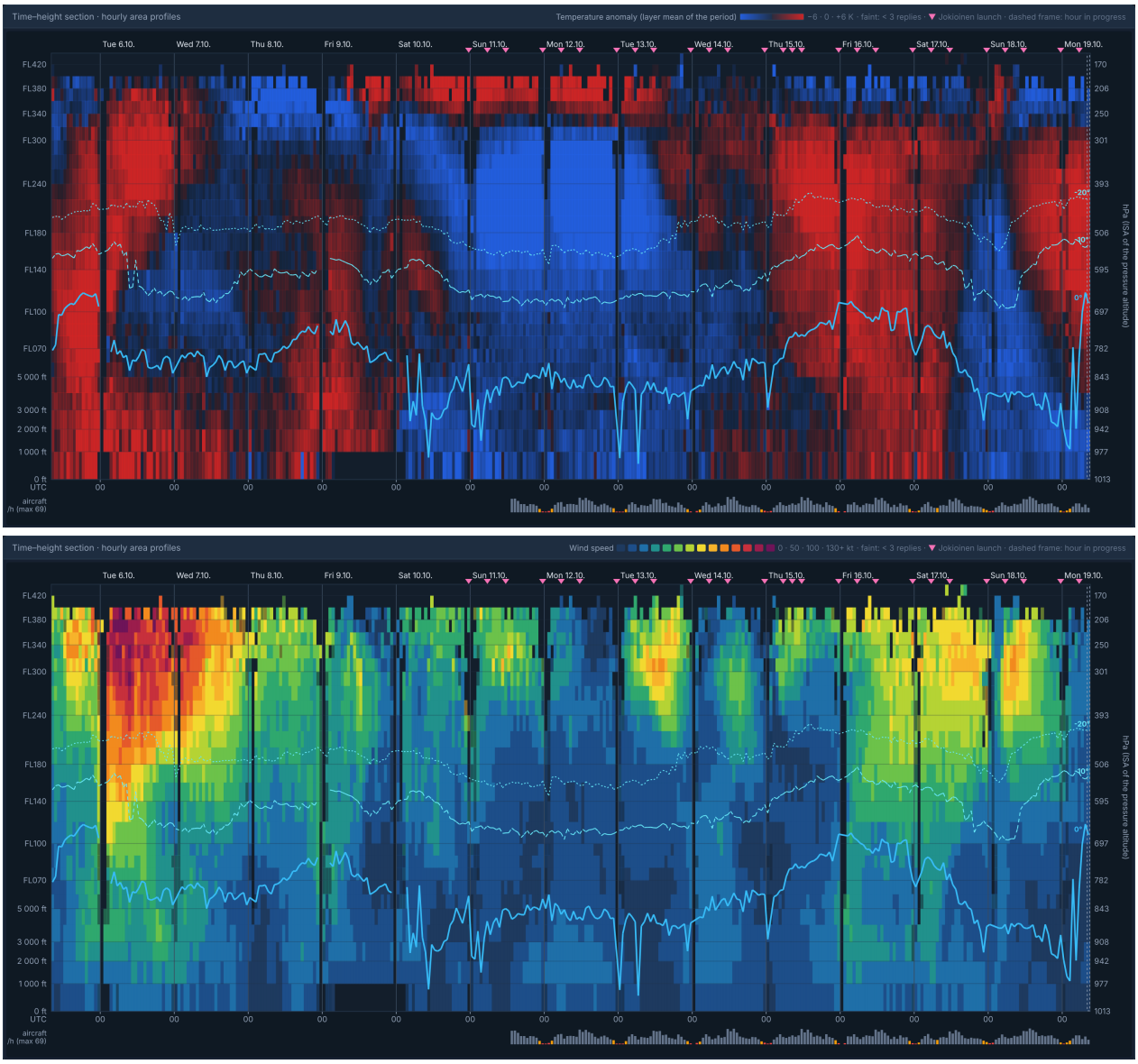



**Figure 21.** 14 days to 19 October 2026, temperature anomaly (top) and wind speed (bottom), barbs off. A jet of up to 145 kt (FL320, 6 October); a cold spell from 10 to 13 October with the 0 °C level near 3 000–5 000 ft and the tropopause low (warm FL340–FL380); warm air again from 14 to 17 October, the front on 17 October and the high of 18–19 October. The dark vertical lines are night hours without data.

### 5.13 A suggested way to use it

1. Start with 72 h, Temperature anomaly, barbs and isotherms on: find the air masses and any sloping boundaries.
2. Switch to Temperature change 3 h to time them, and check the barbs for veering or backing with height.
3. Look at Wind speed for the jet and low-level jets, and at Lapse rate for inversions and steep layers.
4. Use Vertical motion and Vorticity for the dynamics: rising air ahead of troughs and fronts, sinking air under ridges and highs, and the trough passages themselves; Height change 6 h shows the pressure surfaces falling ahead of them at every level.
5. Click an interesting column to open that hour on the Skew-T, with the Indices panel and the radiosonde.
6. Before trusting a single column, check the aircraft strip and the faint cells.

## 6 Sonde check: how good is the aircraft profile?

The Sonde check view compares the Mode S area profile with every stored Jokioinen sounding, from a table that the hourly jobs fill by themselves. The period buttons choose 7, 30, 90 days or a year (Fig. 22).

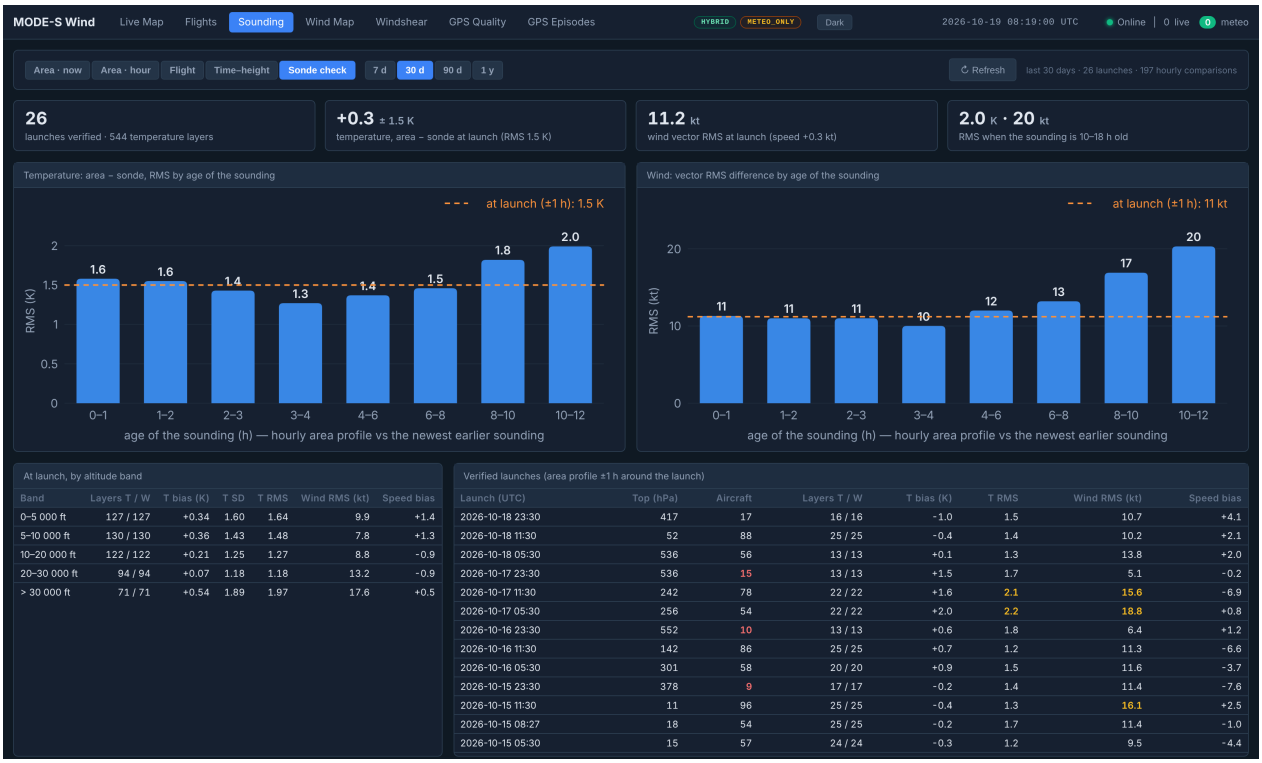



**Figure 22.** Sonde check, 30 days to 19 October 2026: 26 launches, temperature area − sonde $+0.3 \pm 1.5$ K, wind vector RMS difference 11 kt; the difference by the age of the sounding (bars) and by altitude band; the list of launches.

**At launch.** Each sounding is compared with the area profile of $\pm 1$ hour around its launch. For temperature the page gives the *bias* (mean difference area − sonde), the *standard deviation* (scatter around the bias) and the *RMS* difference, $\sqrt{\text{bias}^2 + \text{SD}^2}$; for wind the RMS length of the vector difference and the speed bias. The table by altitude band and the list of launches show where and when the two disagree. Amber values in the list (T RMS $> 2$ K, wind RMS $> 15$ kt) and red aircraft counts ($< 16$, a thin night profile) point to cases worth a look. In October 2026 the two worst launches were 17 October 05:30 and 11:30 UTC, both with the cold front between Jokioinen and Helsinki (Fig. 2) — a difference in the weather, not in the measurement.

**Why they can never agree perfectly.** The two systems measure different things. The radiosonde is a point that drifts with the wind (tens of kilometres by the time it reaches cruise levels), measures every few metres and needs about an hour to reach 10 km. The area profile is a 300 km wide, 60-minute average, 100 km away. Part of the difference is therefore real *representativeness*: inland Jokioinen has colder nights near the ground, fronts and showers pass one site before the other, and jet cores are narrower than 300 km — which is why the wind differences are largest above 30 000 ft. The 30-day temperature bias of about $+0.3$ K is common to the whole fleet; it cannot be found by comparing aircraft with each other, only against an independent reference like this one.

**GNSS height calibration.** Below the comparison tables the view shows the calibration of the GNSS heights of the pressure surfaces (Section 5.8): the correction per 5 000 ft band with the number of airframe cells behind it and the built-in default; the mean offset *per aircraft type*; and the *per-airframe altimeter offsets* — registration and type, the soundings the airframe was compared with, its raw mean offset after the band correction and the offset actually used, which is shrunk towards zero until the airframe has been seen at several soundings (only about 40 % of an airframe's offset repeated from one day to the next in October 2026). On 20–21 October the A320s were on average 12 m above the fleet, the ATR 72s 7.5 m below it and the A321s at 0 m. The aircraft are compared with the balloon at its drifted position, followed from the sounding's own winds.

**By age of the sounding.** Every archived hour is also compared with the newest sounding launched by the middle of that hour. The bar charts give the RMS difference by the age of that sounding: how fast the atmosphere moves away from a sounding. In October 2026 the temperature difference stayed at 1.3–1.6 K for the first 6 hours and grew to 2.0 K at 10–12 hours; the wind difference stayed at 10–12 kt for 6 hours and reached 20 kt at 10–12 hours. The first hours show the floor set by measurement and representativeness; the growth after that is the weather changing. Since soundings are made only every 6 or 12 hours (there is no 18 UTC launch), this is what a live aircraft profile adds: between the launches it follows the atmosphere, while the last sounding grows steadily older.

# 7 The Wind Map page

The Sounding page shows the atmosphere above one place, changing in time. The Wind Map page shows one level of it over the whole area the receiver covers: the wind and temperature measured by the aircraft at a chosen flight level, averaged into square cells and drawn as wind barbs on a map. Around it are the tools of an upper-air chart: isotachs and streamlines, the jet stream, height contours of the pressure surface with the geostrophic wind, the radiosonde, and comparisons with an earlier period or another level.

## 7.1 The page at a glance

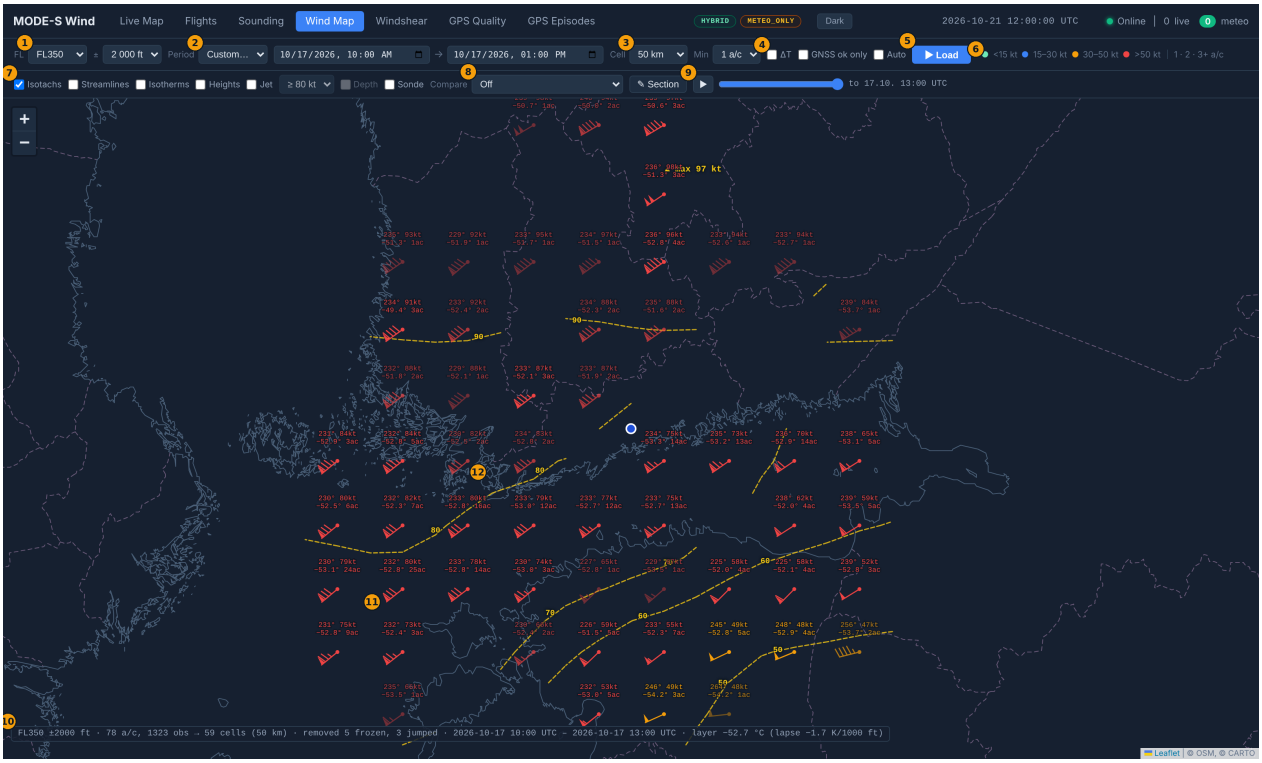



**Figure 23.** The Wind Map, FL350 ±2 000 ft, 17 October 2026 10–13 UTC, isotachs on: 78 aircraft and 1 323 observations in 59 cells of 50 km (5 frozen and 3 jumped positions removed), layer mean −52.7 °C. A south-westerly flow of 230–236° weakens from 90–97 kt in the north-west to 50 kt in the south-east: the jet stream lies to the north-west. (The map background in the screenshots of this section is the coastline and border outline; the page shows the CARTO map.)

**(1) Level** FL chooses the level (1 000 ft to FL450) and ± the depth of the band around it (default ±1 000 ft; ±2 000 ft gives more cells).

**(2) Period** the last 1, 3, 6, 12 or 24 hours, or Custom with a start and end.

**(3) Cell and Min** the cell size (25, 50 or 100 km) and the minimum number of different aircraft in a cell for it to be shown.

**(4) $\Delta T$, GNSS ok only, Auto** temperature colouring (Section 7.3); leave out aircraft while their GPS is degraded; reload every 2 minutes.

**(5) Load** runs the query (also Enter); the page loads at once when opened, with the remembered settings.

**(6) Legend** barb colour by speed ($<15$, 15–30, 30–50, $>50$ kt) and barb opacity by the number of aircraft (1, 2, 3+).

**(7) Analysis** Isotachs, Streamlines, Isotherms, Heights, Jet with its speed and Depth, Sonde; Section draws a vertical cross-section (Section 7.9).

**(8) Compare** change since the previous period, or the difference to another level (thermal wind).

**(9) Time slider** moves the period back hour by hour; ▶ animates the last 24 hours.

**(10) Status line** aircraft and observations used, cells shown, positions removed, period, layer mean temperature and lapse rate; QNH for low levels; the geostrophic wind with Heights.

**(11) Cell** a barb with a two-line label: direction and speed, temperature at the level and number of aircraft. Click for the popup, hover for a short tooltip.

**(12) Isotach** dashed yellow lines of equal speed every 10 kt; `max 97 kt` marks the strongest analysed wind.

### 7.2 How the map is made

**Cells and averaging.** The observations of the band and period are placed on a grid of square cells on a flat map centred on the receiver — square on the ground, not in degrees (a degree grid would be twice as tall as wide at 60°N). Within a cell each aircraft's observations are averaged first (wind as $u$, $v$ components, so 350° and 010° give 000°), then the aircraft means are averaged: one aircraft with 30 stored replies counts as much as one with 3. The barb's opacity shows how many different aircraft are behind it — faint for one, medium for two, full for three or more — because a single airframe's heading or temperature error is not averaged out in a faint cell. The popup (Fig. 24) gives the counts and the spread between the aircraft.

**Positions.** Under GNSS interference an aircraft's position can freeze (the last fix is repeated while it moves on) or jump to a false, spoofed place — in this area often a fixed point east of St Petersburg. The wind itself is not affected, but it would land in the wrong cell, so frozen positions (the same position $\geq 10$ s later while moving) and impossible jumps (more than 3 NM plus 800 kt times the elapsed time) are removed. GNSS ok only also leaves out all observations while the aircraft had a GPS degradation episode; it is off by default, because at cruise levels about two thirds of the observations fall inside an episode while most positions still move normally (inertial navigation).

**Temperature at the level.** A band of ±2 000 ft spans 6 K or more of normal cooling with height, more than the horizontal differences. Each temperature is therefore moved to the band centre with the lapse rate fitted to all temperatures of the band and period (shown in the status line).

**Low levels.** Transponders send pressure altitude. Below FL050 the band is shifted to the pressure altitude of the chosen height above mean sea level with the QNH of the period — the current METAR for a recent period, otherwise the METAR stored with the hourly profile archive — shown in the status line.

**How many cells to expect.** With 50 km cells, ±2 000 ft and 3 hours: at FL350 about 50 cells, of which about 25 with three or more aircraft (17 at night); at FL150–FL250 about 9 such cells, mostly within 100 km of Helsinki; at FL050 and below only the 3–4 cells around the airport — for the lowest levels the airport low-level profile of the Skew-T is the better tool. At cruise levels the cells reach about 250–300 km from the receiver.

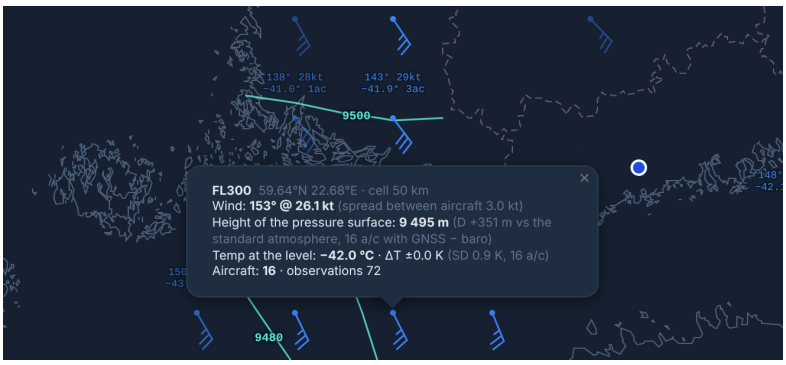



**Figure 24.** Popup of one cell, FL300, 20 October 2026 09–12 UTC with Heights: wind 153°/26 kt with a spread of 3 kt between 16 aircraft; height of the 301 hPa surface 9 495 m (D +351 m); −42.0 °C.

### 7.3 $\Delta T$: warm and cold air at the level

With $\Delta T$ each cell is tinted by its temperature minus the mean of all cells of the layer (red warmer, blue colder, full at ±3 K). It removes the level's normal temperature and leaves the air masses: warm and cold advection areas, frontal zones, and near the tropopause the stratospheric air on the cold side of the jet.

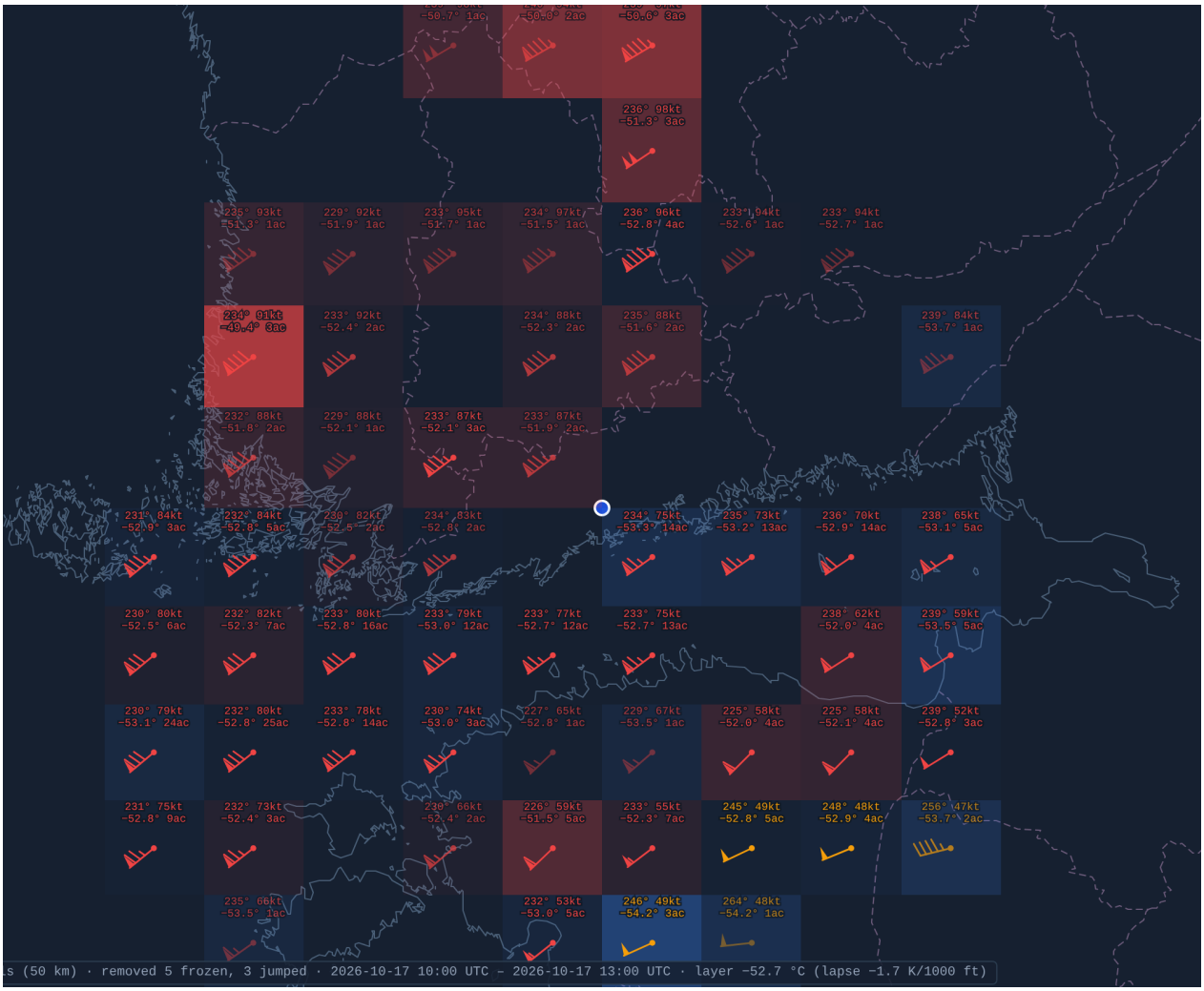



**Figure 25.** $\Delta T$, same level and period as Fig. 23. At FL350 the cells north-west of Helsinki are 1–3 K warmer (−49 to −52 °C) than those to the south-east (−52 to −54 °C). Near the tropopause this is typical for the cold (poleward) side of a jet: there the tropopause is lower, and FL350 is already in the warmer stratosphere.

### 7.4 Isotachs and streamlines

For these the cell winds are smoothed onto a grid of points half a cell apart (a Barnes-type average with a Gaussian weight of one cell length, each cell weighted by its number of aircraft, up to 10). A grid point is analysed only when a cell centre lies within $\frac{3}{4}$ of a cell, so the lines are drawn only where there are data, never extrapolated. **Isotachs** are dashed yellow lines every 10 kt (20 kt when the speeds span more than 60 kt); `max` marks the strongest analysed wind, which may lie at the edge of the data. **Streamlines** are traced through the smoothed wind with a small arrow showing the flow. **Isotherms** — dotted red lines of equal temperature at the level, every 1 K (2 K when the range exceeds 8 K) — come from the same smoothed field, with the band's temperatures moved to its centre (Fig. 27). Wind blowing across the isotherms from warm to cold air is warm advection, from cold to warm cold advection. All of them follow the Min filter.

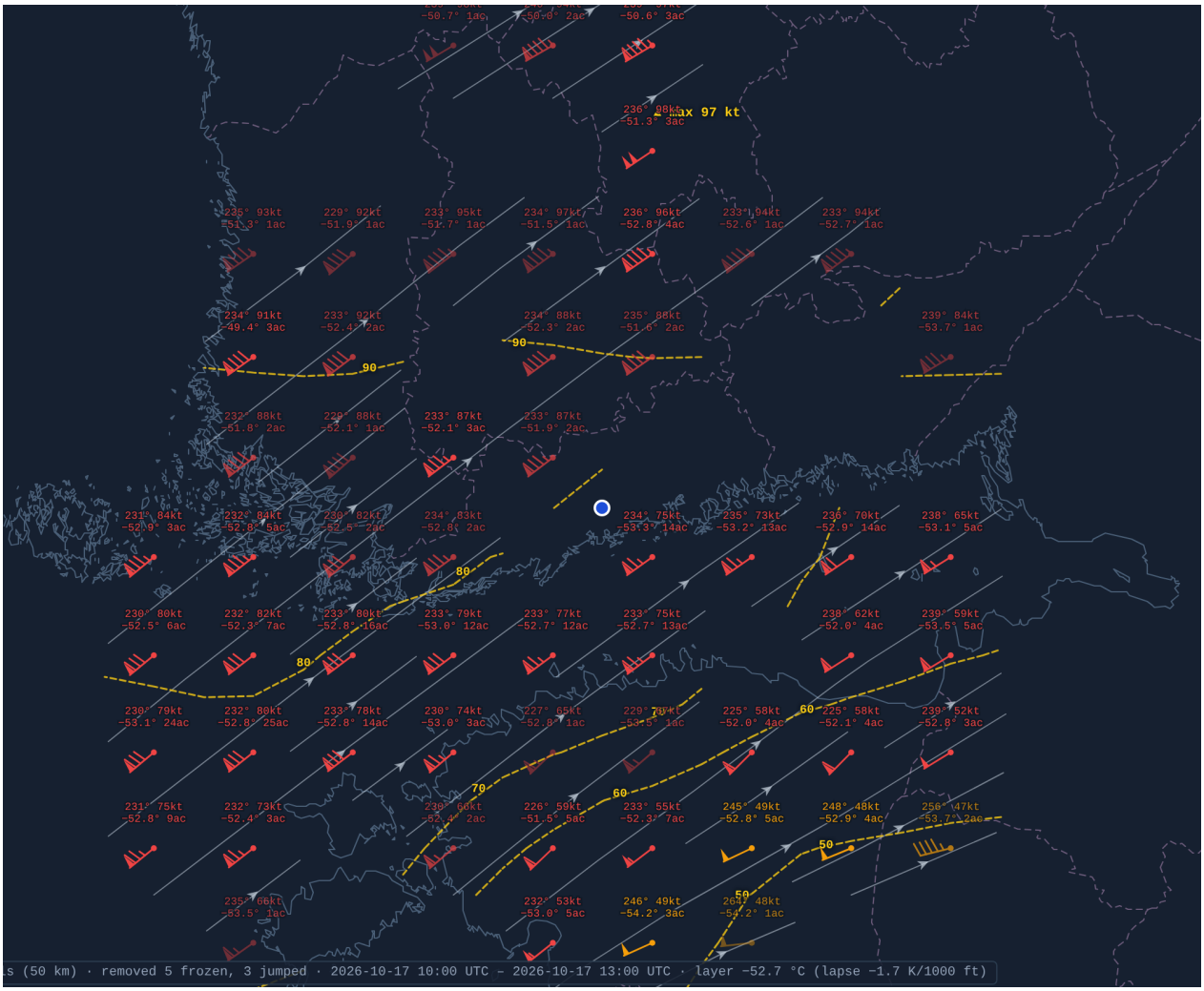



**Figure 26.** Isotachs and streamlines, FL350, 17 October 2026 10–13 UTC. Straight south-westerly flow; the isotachs run along it, from 90 kt in the north-west to 50 kt in the south-east — the right-hand (warm, south-eastern) flank of a jet whose core lies beyond the north-western edge of the data.

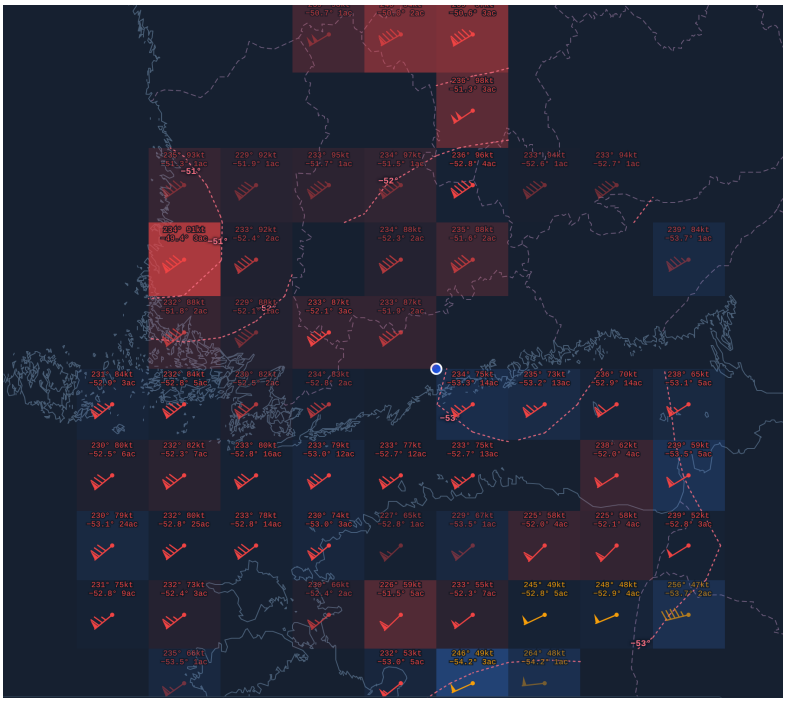


**Figure 27.** Isotherms with $\Delta T$, FL350, 17 October 2026 10–13 UTC: −51 °C in the north-west, −52 to −53 °C towards the south-east — the same pattern as in Fig. 25.

### 7.5 Jet stream

With Jet the jet stream is drawn as on aviation significant-weather (SIGWX) charts: a thick line along its **axis** with an **arrowhead** at the downstream end, **barb flags** on the line at the maximum (pennant 50 kt, flag 10 kt, half flag 5 kt) with the level and speed, and double **change bars** across the line where the speed has fallen by 20 kt, 40 kt … from the maximum. The selector next to it chooses the speed that counts as jet: ≥ 80 kt as on aviation charts, or ≥ 60 kt for weaker jets.

**How the axis is found.** In the smoothed field (as for the isotachs) the mean direction of the points with at least the chosen speed defines the flow. The field is cut into sections across the flow, one cell apart, and in each section the strongest point lies on the axis. The sections with at least the chosen speed around the overall maximum form the jet (one empty section is bridged), and the axis is smoothed over three sections.

**Solid or dashed.** The line is **solid** where the speed falls off on both sides of the axis inside the data — there the axis is really seen. It is **dashed** where the strongest wind of a section lies at the edge of the data, so the real axis is probably beyond what the receiver sees; the label then adds ↑ `stronger beyond?`. At cruise levels the data reach about 250–300 km from Helsinki and a jet is often wider than that: in the jet of 16–18 October 2026 (85–114 kt) the maximum lay at the edge of the data in almost every 3-hour period. The direction, speed and timing of the jet are then right, but its exact position is not.

**Depth.** With Depth the levels FL250, FL290, FL330, FL370 and FL410 (±2 000 ft) are loaded as well, and the label adds the levels where the wind at the jet maximum is at least the chosen speed, as on the charts: `330/410` means from about FL330 to FL410. It makes loading slower, so it is a separate option. The hover gives the speed at each level.

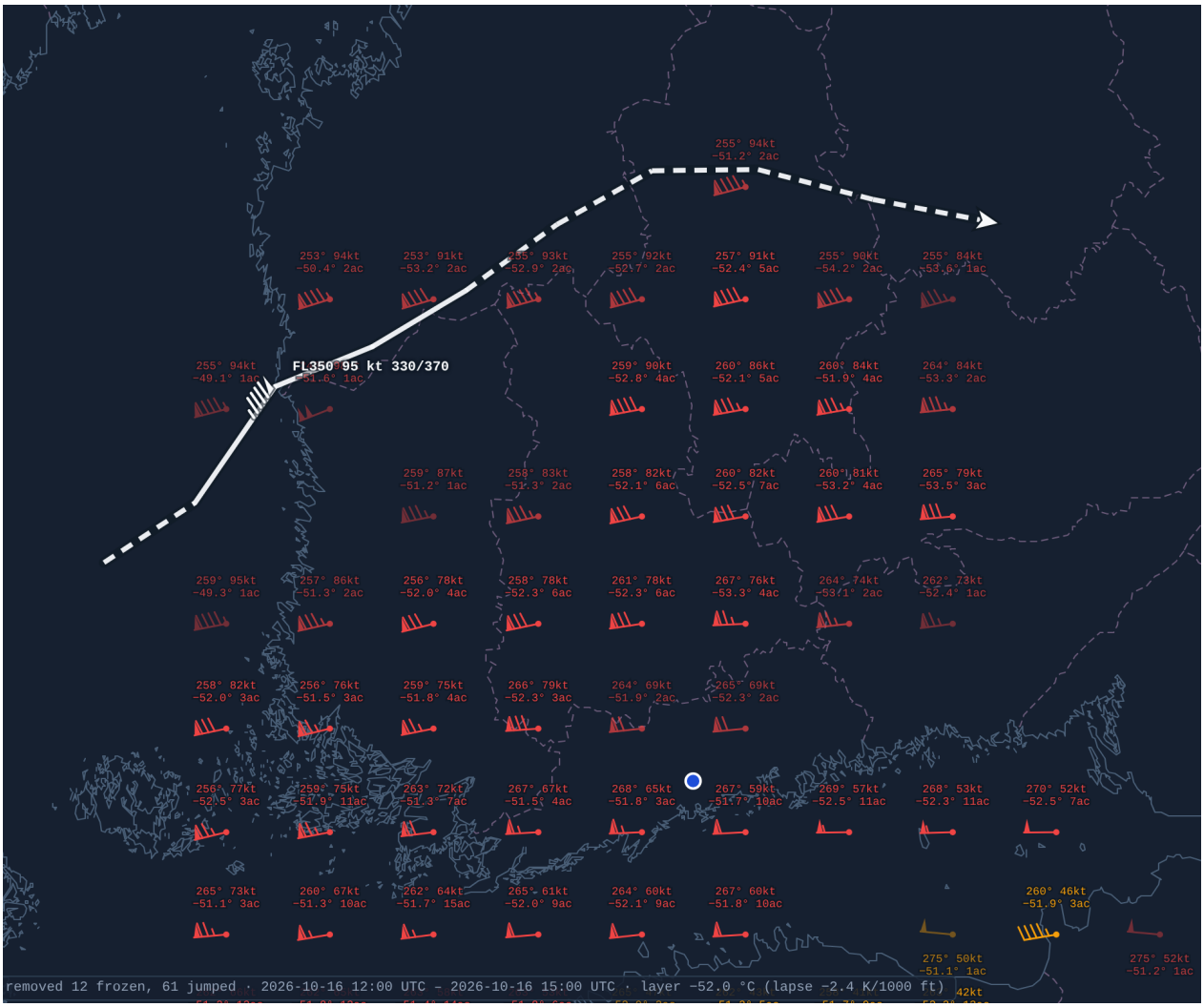



**Figure 28.** Jet, FL350, 16 October 2026 12–15 UTC, with Depth: a west-south-westerly jet of 95 kt, at least 80 kt from about FL330 to FL370. The axis is solid north-west of the receiver, over the west coast, where the speed falls off on both sides inside the data, and dashed at its ends at the edge of the data. The map has been moved north-west to show the whole jet.

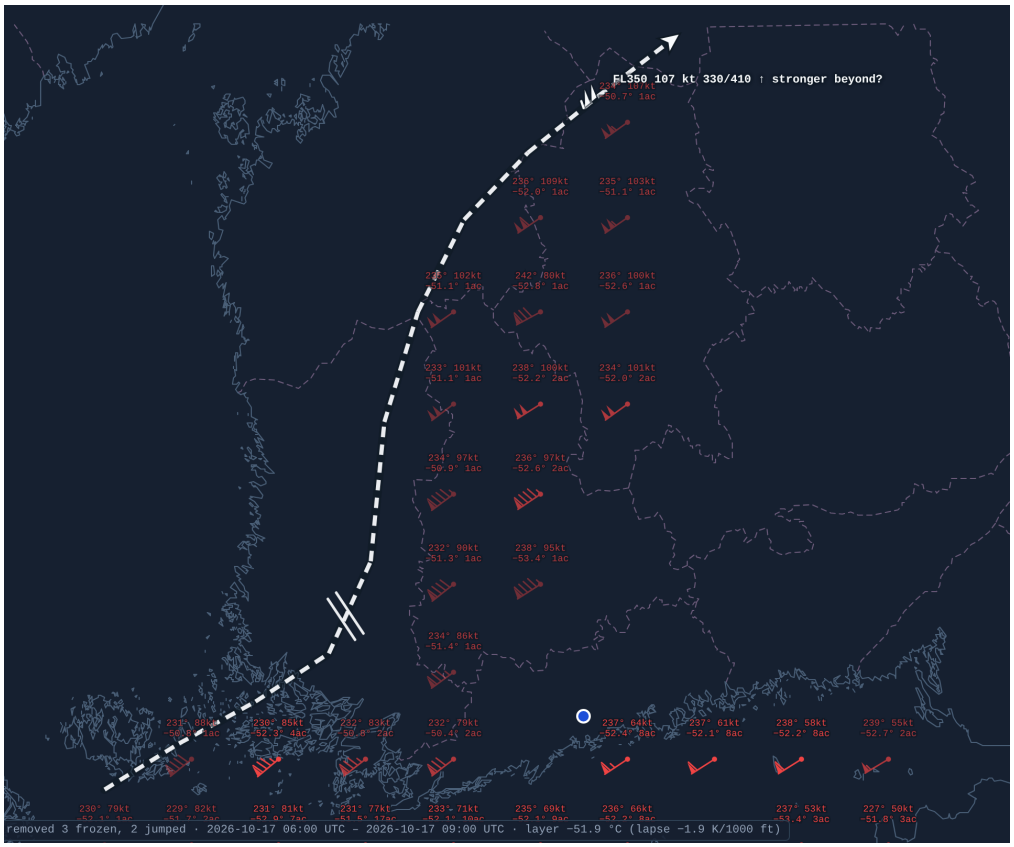



**Figure 29.** Jet, FL350, 17 October 2026 06–09 UTC: 107 kt with ≥ 80 kt from FL330 to FL410. The strongest winds lie along the north-western edge of the data all the way, so the axis is dashed and labelled ↑ `stronger beyond?` — the core of this jet passed north-west of the receiver's coverage. The double bars mark where the speed along the axis has fallen to 87 kt.

### 7.6 Heights and the geostrophic wind

With Heights each cell also gets the height of the pressure surface from the aircraft's GNSS height (with the calibration of Section 5.8; NACp ≥ 8, outside GPS degradation episodes, moved to the band centre with the band's temperature), the median of the cell's aircraft. The heights are smoothed like the winds and drawn as teal **height contours** in metres above mean sea level, every 5, 10 or 20 m depending on the range, only near cells with heights. This is the familiar upper-air chart: away from the ground the wind blows along the contours with the lower heights on its left, and faster where they are closer together.

A plane fitted to all aircraft heights of the band (one point per aircraft and cell; outliers beyond 3 times the RMS, at least 20 m, left out; at least 6 points spread over 30 km in both directions) gives the slope of the pressure surface and from it the **geostrophic wind** of the area,

$$\vec{V}_g = \frac{g}{f}\,\hat{k} \times \nabla z,$$

drawn as a dashed teal arrow `Vg` at the centre of the points, next to the mean measured wind of the same aircraft (`V`, grey). A slope of 10 m per 100 km gives about 15 kt. The tooltip and the status line give both winds, the slope, the *ageostrophic* difference and the residual of the fit. The real wind differs from the geostrophic one where the flow curves — in a high it is somewhat stronger, in a low weaker — and where it accelerates into or out of a jet stream; ageostrophic winds of 5–10 kt are normal.

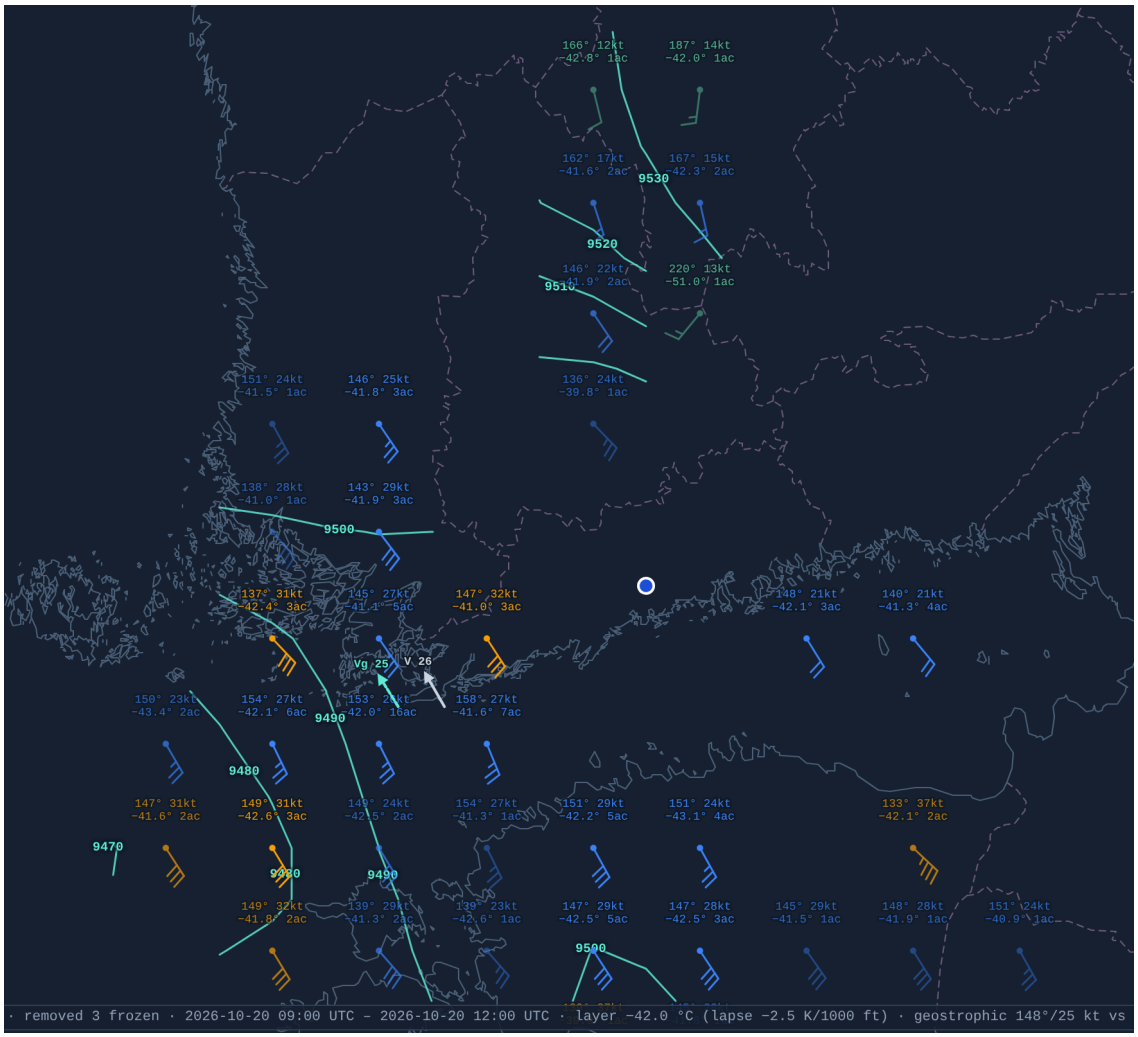



**Figure 30.** Heights, FL300 ±2 000 ft, 20 October 2026 09–12 UTC, under a weakening high: the 301 hPa surface at 9 470–9 530 m (D about +330 to +390 m), the south-south-easterly wind of 20–30 kt blowing along the contours. Geostrophic wind 148°/25 kt against a measured 150°/26 kt from 79 aircraft positions.

**How good is it?** In the first test (20–21 October 2026, a weak high) the geostrophic wind at FL250–FL350 agreed with the measured one within about 10–20° and 2–10 kt, and was poorer at FL180 and below, where the traffic is concentrated near Helsinki and the plane cannot be fitted reliably. The heights themselves need the calibration, which improves with every sounding; this part of the page is experimental and will be tuned as data in stronger flow accumulate.

### 7.7 Radiosonde at the level

With Sonde, the newest Jokioinen sounding launched before the end of the period (at most 24 h old) is drawn at its launch site as a pink barb with a square dot, interpolated to the pressure of the level. The label gives the launch hour, speed and temperature; the popup the values, the age of the sounding and the difference from the nearest Mode S cell. If the sounding (often only its open-data part) stops below the level, a pink ring says so.

### 7.8 Compare: change in time and the thermal wind

**Change since previous period** loads the previous period of the same length (for 12–15 UTC: 09–12 UTC) and draws in every cell with data in both a purple arrow of the *vector change* of the wind, with its size and the temperature change; the cells are tinted red where the level has warmed and blue where it has cooled (full at ±3 K). A front or a trough passes as a band of cooling with a turn of the wind.

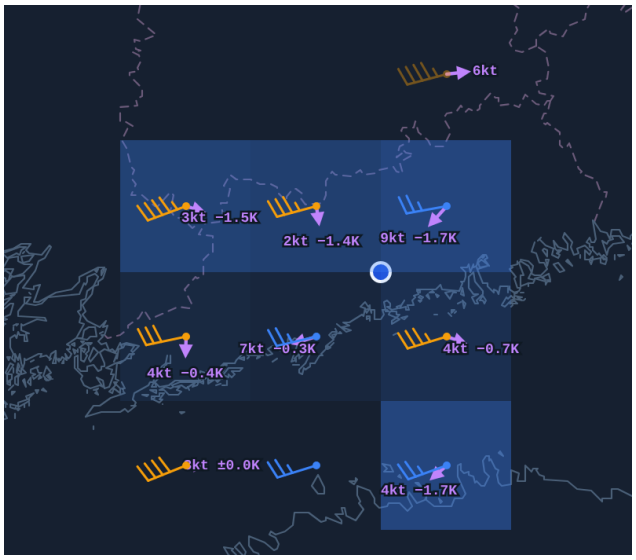



**Figure 31.** Change since the previous period, FL100, 17 October 2026 12–15 against 09–12 UTC: the cells around Helsinki have cooled by 0.4–1.7 K with small changes of the wind (2–9 kt) — the start of the cooling at this level behind the cold front, which the time–height section shows between 12 and 18 UTC (Section 5.2).

**Other level (thermal wind)** loads a second level for the same period and draws in each common cell the *thermal wind*, the vector difference of the wind from the lower to the upper level. It blows along the isotherms of the layer between them with the cold air on its left (Section 5.4). The cells are tinted by how the wind turns with height: **veering** (clockwise) is warm advection, red; **backing** is cold advection, blue (full at 30°, shown when both winds are at least 8 kt). The popup adds the turning and the lapse rate between the levels. Mid-level cells lie mostly near Helsinki, so the thermal wind is usually available within about 100 km.

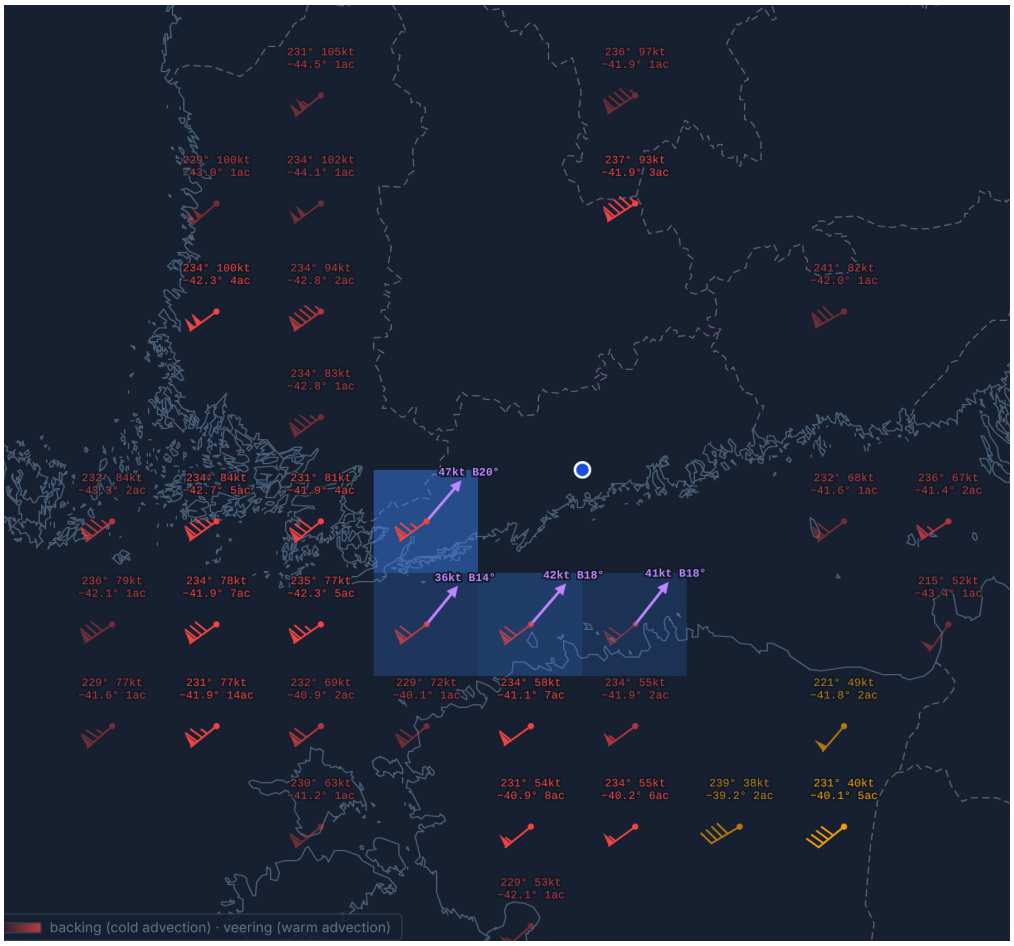



**Figure 32.** Thermal wind FL100 → FL300, 17 October 2026 10–13 UTC: in the four common cells south-west of Helsinki 36–47 kt towards the north-east, with the wind backing 14–20° with height — cold advection in the layer, the cold front of Section 5.2 approaching.

While a comparison is shown, $\Delta T$ is switched off (the tint shows the comparison).

### 7.9 Vertical cross-section along a line

Section puts the map in drawing mode: click the start point A and the end point B (Esc cancels). A panel at the bottom of the map then shows a vertical cross-section along the line for the map's period (Fig. 33): all observations within one cell size of the line, at all levels, binned every half cell along it and into the time–height layers, each aircraft averaged first, with the same position checks as the map. Distance from A runs across, pressure altitude up (square-root scale, ISA pressure on the right). The colour field can be the **temperature anomaly** (minus the layer's mean along the section, full at ±4 K — fronts slope through it), the **wind speed**, the **wind across the section** (red blowing from the left to the right looking from A to B, blue the other way, full at ±60 kt — a jet crossing the line shows as a core) or the **height of the pressure surface** (D-value). Barbs and the 0, −10 and −20 °C isotherms are drawn as in the time–height section; faint cells rest on one aircraft, and the tooltip gives the values. The section follows the period when the map is reloaded; × closes it. Low levels have data only near the airport, so a long section mostly shows the middle and upper troposphere.

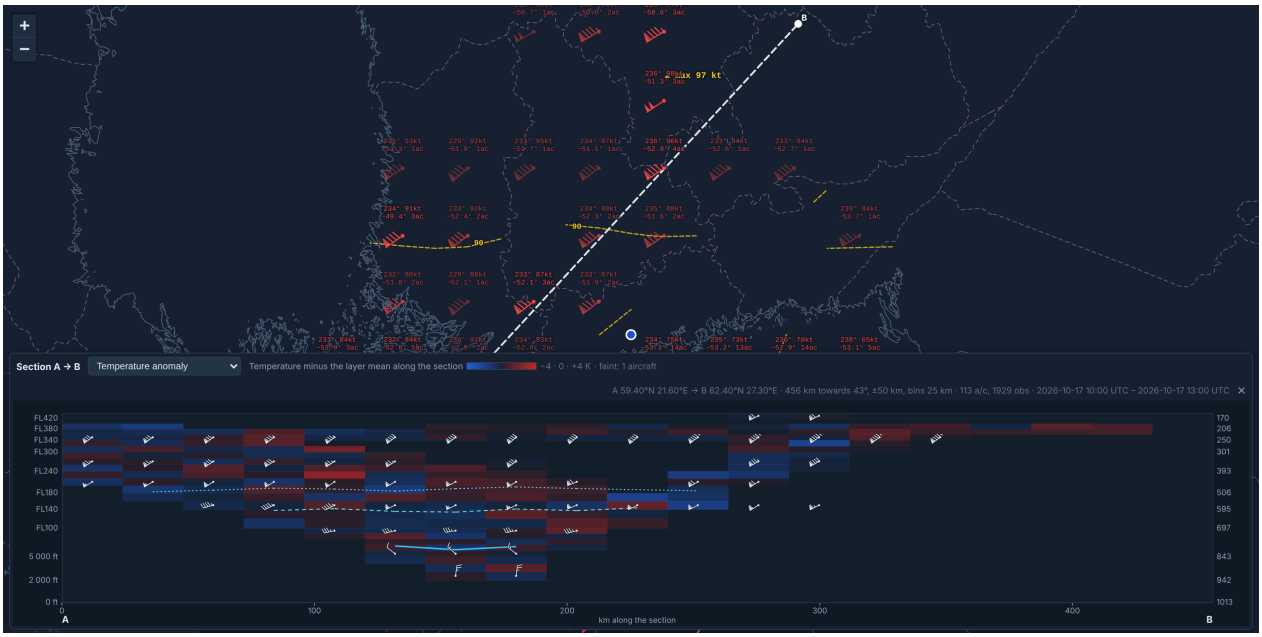



**Figure 33.** Section from the Turku archipelago (A) to eastern Finland (B), 456 km towards 043°, on the FL350 map of 17 October 2026 10–13 UTC; 113 aircraft within ±50 km. Temperature anomaly with barbs: the strong south-westerly flow at FL300–FL380 along the whole line, the −20 and −10 °C levels near FL190 and FL140, the 0 °C level near 6 000 ft over Helsinki.

### 7.10 Time slider, animation and settings

The slider moves the period back in whole hours, up to 23 hours before the chosen end; the label shows the new end. ▶ steps through the last 24 hours (about one frame a second, wrapping around) and ■ stops. On the first slider move or ▶ all 24 frames are loaded at once and kept, so the slider and the animation show them without waiting (a thin strip under the slider marks the loaded frames). With a Last … period the earlier positions end on whole hours (−1 = the last full hour). Auto reloads every 2 minutes for the Last … periods when the slider is at the end, and at once when the page becomes visible again. All settings, including a custom period, are remembered in the browser; the map follows the page's Light / Dark theme.

### 7.11 A suggested way to use it

1. Start at FL300–FL350 with 3 hours, ±2 000 ft and 50 km cells: cruise levels have the most aircraft. Switch on Isotachs and Jet to see where the strongest winds are and whether the jet core is inside the data.
2. Use $\Delta T$ for the air masses and Streamlines for troughs and ridges; check the faint cells before reading anything into a single barb.
3. With Heights, compare the measured wind with the geostrophic one and see the slope of the pressure surface.
4. Use Compare to see what has changed in the last hours, and the thermal wind between FL100 and FL300 for warm and cold advection.
5. Run the animation over the last 24 hours to watch a front or a jet pass, and open the same hours in the time–height section of the Sounding page for the vertical view.

## 8 Limitations

**Where aircraft fly.** The profile exists only where aircraft fly: cruise levels are well sampled by day, the lowest levels mainly along the EFHK approach and departure paths, and the night hours (about 22–02 UTC) may have only a few aircraft. Watch the data badge, the red rings, the A/C column and the aircraft strip of the time–height section. Data usually end at about FL410, so a high tropopause is missed.

**Averaging.** The area profile is an average over 150 km and 60 minutes. A front passing through the area is smeared over several hours, thin inversions and a sharp tropopause are smoothed over a 1 000–2 000 ft layer, and jet maxima are weaker than in a point measurement.

**Accuracy.** Layer mean temperatures are good to about 1–1.5 K against the sonde, the lowest layers less certain because single replies at approach speeds scatter by 3–4 K; winds are good to about 10 kt RMS. Single-flight profiles carry the scatter of one airframe.

**No humidity aloft.** Aircraft report no humidity, so indices that need dewpoints aloft are sonde only; the moisture trial of

Section 5.9 gives only a rough dry / moist / humid indication at 1 000–15 000 ft near the airports. The surface parcel uses the METAR dewpoint and assumes a well-mixed boundary layer; clouds above the ground, icing itself and contrail persistence cannot be derived.

**Heights.** The vertical scale is pressure altitude (ISA). The Indices panel computes true heights hypsometrically from the METAR surface pressure; plain temperature is used throughout (no virtual-temperature correction), which changes the 1000–500 hPa thickness by well under 1 dam in Finnish autumn air.

**Derived fields.** Vertical motion and vorticity are derived from averaged temperatures and winds and inherit their errors, multiplied: vertical motion by the division by $\Gamma_d - \Gamma$, vorticity by the slope of a plane fitted to a few dozen aircraft. Both are drawn only where they were tested to work (vertical motion 5 000 ft–FL200, vorticity where enough aircraft are spread over the area), and both are faint where they rest on little data.

**GNSS heights.** The heights of the pressure surfaces, the thickness temperatures and the geostrophic wind rest on the calibration against the Jokioinen soundings, which corrects the aircraft's ellipsoidal GNSS height and speed-dependent pressure error; it is new and improves with every sounding. GNSS jamming removes most aircraft at night, and the geostrophic wind needs aircraft spread over the area (in practice FL250 and above).

**Wind Map coverage.** The map shows only where aircraft fly within the receiver's range — at cruise levels about 250–300 km, at mid and low levels much less. Jet cores and fronts often lie partly outside; the dashed jet axis and the `max` label at the edge of the data say so. A longer period or a wider band fills more cells but mixes weather that moved meanwhile.

**Not an official product.** The automatic indices and analysis follow simple, documented rules on averaged data. Use them to understand the atmosphere, not for flight planning; use official forecasts and soundings for that.

# 9 Quick reference

| Item | Meaning |
|---|---|
| *Time–height section* | |
| red / blue (anomaly) | warmer / colder than the layer's mean over the shown period (full at $\pm 6$ K) |
| red / blue (3-hour change) | warming / cooling over the last 3 hours (full at $\pm 4$ K) |
| blue → green → yellow → red → purple | wind speed in 10 kt steps (purple $\geq 130$ kt) |
| violet / dark blue / light blue / grey / orange / red | lapse rate $< -1$ / $-1$–2 / 2–5 / 5–7 / 7–9 / $\geq 9$ K/km |
| solid / dashed / dotted cyan line | 0 / $-10$ / $-20$ °C (lowest crossing per hour) |
| faint cell, dark cell | fewer than 3 replies, no data |
| pink triangle | Jokioinen radiosonde launch |
| red / amber bar in the strip | fewer than 8 / 20 aircraft in the hour |
| green / orange (vertical motion) | rising / sinking air (full at $\pm 3$ cm/s, 5 000 ft–FL200) |
| purple / gold (vorticity) | cyclonic / anticyclonic (full at $\pm 15 \times 10^{-5}$ s$^{-1}$; faint: error $> 4$ or $< 10$ aircraft) |
| red / blue (height of the pressure surfaces) | higher / lower than in the standard atmosphere (D-value, full at $\pm 300$ m) |
| red / blue (height change 6 h) | surfaces rising / falling (full at $\pm 60$ m) |
| red / blue (thickness check) | thickness temperature warmer / colder than Mach/TAS (full at $\pm 2$ K) |
| brown / green / blue (moisture, trial) | dry / moist / humid: index $< 40$ / 40–70 / $\geq 70$ % (about RH), 1 000–15 000 ft; faint: uncertain |
| magenta dashed `TP`; yellow `Vmax` | tropopause; height of the strongest wind ($\geq 50$ kt) |
| narrow pink-framed column | a Jokioinen sounding in the same colour field |
| dashed frame | hour in progress (computed live) |
| *Wind Map* | |
| barb colour; opacity | $< 15$ / 15–30 / 30–50 / $> 50$ kt; 1 / 2 / 3+ aircraft in the cell |
| cell label | direction and speed / temperature at the level and aircraft |
| red / blue tint ($\Delta T$) | warmer / colder than the layer mean (full at $\pm 3$ K) |
| dashed yellow lines, `max` | isotachs every 10 (20) kt; strongest analysed wind |
| thin grey lines with arrows | streamlines |
| dotted red lines | isotherms at the level (1 or 2 K) |
| section panel colours | temperature anomaly / wind speed / wind across (red left→right) / D-value |
| thick line with arrowhead, flags | jet axis ($\geq 80$ or 60 kt), maximum speed; dashed: maximum at the edge of the data |
| double bars on the jet | speed 20, 40 … kt below the maximum |
| `330/410` | jet depth: levels with $\geq$ the jet speed at the maximum |
| teal lines; `Vg`, `V` | height contours (m MSL); geostrophic and measured mean wind |
| pink barb with square dot; pink ring | Jokioinen sonde at the level; sounding ends below it |
| purple arrows, red / blue tint (compare) | wind change and warming / cooling; thermal wind and veering / backing |
| *Indices panel (amber / red)* | |
| CAPE | $\geq 100$ / $\geq 1\,000$ J/kg |
| Lifted Index; Showalter | $\leq -1$ / $\leq -5$; $\leq 0$ / $\leq -3$ |
| K index; Total Totals; Vertical Totals | $\geq 26$ / $\geq 31$; $\geq 44$ / $\geq 50$; $\geq 30$ |
| lapse rates | $\geq 7$ / $\geq 9$ K/km |
| shear 0–1 km; 0–6 km | $\geq 20$ kt; $\geq 40$ kt (amber) |
| CCL · T conv amber | the surface has reached the convective temperature |
| *Level table and Sonde check* | |
| red A/C | layer measured by only 1–2 aircraft |
| amber $\Delta T$; $\Delta V$ | more than 2 K; more than 10 kt from the sonde |
| amber T RMS; wind RMS; red aircraft | $> 2$ K; $> 15$ kt; $< 16$ aircraft behind a launch comparison |

# A Formulas

- Mach/TAS temperature: $T = 288.15\,(\mathrm{TAS}/(661.47\,M))^2$ K (TAS in kt).
- ISA pressure of pressure altitude $h$ (ft): $p = 1013.25\,(1 - 6.8756 \times 10^{-6}\,h)^{5.2559}$ hPa below 36 089 ft, $p = 226.32\,\exp\big(-(h - 36\,089)/20\,806\big)$ above.
- Hypsometric thickness: $\Delta z = \dfrac{R\bar{T}}{g}\,\ln\dfrac{p_1}{p_2}$, $R = 287.04$ J/(kg K), $g = 9.80665$ m/s$^2$. For 1000–500 hPa: $\bar{T} = g\,\Delta z/(R\ln 2)$, so 1 dam $\approx 0.49$ K.
- Lapse rate between two levels: $\Gamma = -(T_2 - T_1)/(z_2 - z_1)$; time–height: central difference between the layers below and above, with temperatures smoothed over $\pm 1$ h (weights 0.5 / 1 / 0.5 times the number of replies).
- Anomaly: $T'(t,z) = T(t,z) - \overline{T}(z)$, the mean over the shown period weighted by replies; 3-hour change: $T(t,z) - T(t - 3\,\mathrm{h}, z)$.
- Local temperature change: $\partial T/\partial t = -\vec{V}\cdot\nabla T + (\Gamma_d - \Gamma)\,w + Q$, $\Gamma_d = g/c_p = 9.8$ K/km.
- Geostrophic wind: $\vec{V}_g = \dfrac{g}{f}\,\hat{k}\times\nabla_p z$, $f = 2\Omega\sin\varphi \approx 1.26 \times 10^{-4}$ s$^{-1}$ at 60°N.
- Thermal wind: $\vec{V}_T = \vec{V}_g(p_2) - \vec{V}_g(p_1) = \dfrac{R}{f}\,\ln\dfrac{p_1}{p_2}\,\hat{k}\times\nabla\bar{T}$; thermal advection $-\vec{V}\cdot\nabla\bar{T} = -\dfrac{f}{R\ln(p_1/p_2)}\,(u_1 v_2 - v_1 u_2)$ (positive = warm advection, wind veering with height).
- LCL (Bolton 1980): $T_L = 1\big/\big(1/(T_d - 56) + \ln(T/T_d)/800\big) + 56$ (K), $p_L = p_0\,(T_L/T)^{c_p/R}$.
- CAPE $= R\int_{p_{EL}}^{p_{LFC}} (T_p - T_e)\,d\ln p$; CIN the negative part below the LFC; Lifted Index $= T_e(500) - T_p(500)$; Showalter the same for a parcel from 850 hPa.
- K index $= (T_{850} - T_{500}) + T_{d,850} - (T_{700} - T_{d,700})$; Vertical Totals $= T_{850} - T_{500}$; Total Totals $=$ VT $+ (T_{d,850} - T_{500})$.
- Precipitable water: $W = \dfrac{1}{g}\int_{p_{300}}^{p_s} r\,dp$, $r$ the mixing ratio.

- Vertical motion (adiabatic method): $w = \dfrac{\mathrm{ADV} - \partial T/\partial t}{\Gamma_d - \Gamma}$ with $\mathrm{ADV} = -\dfrac{f}{R\ln(p_1/p_2)}\,(u_1 v_2 - v_1 u_2)$ between the layers two below (1) and two above (2), $\partial T/\partial t = (T_{+3\,\mathrm{h}} - T_{-3\,\mathrm{h}})/6\,\mathrm{h}$, $\Gamma$ from the layers next to it; not computed where $\Gamma_d - \Gamma < 1.5$ K/km.
- Vorticity and divergence: least-squares plane $u = a_u + b_u x + c_u y$, $v = a_v + b_v x + c_v y$ through one point per aircraft; $\zeta = b_v - c_u$, $\delta = b_u + c_v$; standard errors from the residual variances and $(X^{\mathsf{T}}X)^{-1}$; 1 kt/km $= 51.4 \times 10^{-5}$ s$^{-1}$.
- GNSS height of a pressure surface: $z = 0.3048\,(h_p + \Delta h_{\mathrm{GNSS-baro}}) - c(h_p) - b_{\mathrm{airframe}}$ (m above mean sea level), $h_p$ the pressure altitude (ft); $c$ per 5 000 ft band (linear between band centres) and $b$ fitted against the Jokioinen soundings of the last 30 days. D-value $D = z - 0.3048\,h_p$.
- Thickness temperature: $\bar{T}_v = g_0\,(H_2 - H_1)/(R\ln(p_1/p_2))$ with geopotential heights $H = (g_\varphi/g_0)\,R_E z/(R_E + z)$, $R_E = 6\,371$ km.
- Moisture index (trial): $100\,(T_v - T - b)/(0.608\,q_{\mathrm{sat}}(T,p)\,T)$, $T_v - T \approx 0.608\,qT$; $b$ the sonde-learnt offset per layer and climb / descent ($\sum d_i/(n+3)$ over 30 days); $q_{\mathrm{sat}}$ from Bolton's $e_s = 6.112\exp(17.67t/(t+243.5))$ hPa, $q = 0.622\,e/(p - 0.378\,e)$.
- Geostrophic wind of an area: plane $D = a + bx + cy$ through the aircraft heights; $u_g = -(g/f)\,c$, $v_g = (g/f)\,b$; 10 m per 100 km $\approx$ 15 kt at 60°N.
- Jet axis: in sections across the mean flow of the points with $\geq$ the jet speed (one cell apart), the strongest point of the smoothed field; "edge" when that point lies within $\frac{3}{4}$ grid step of the section's data edge.
- WMO tropopause: the lowest level above 5 km at which the lapse rate falls to $\leq 2$ K/km, provided the average lapse rate to every level within the next 2 km does not exceed 2 K/km.
- Verification: bias $= \overline{\Delta T}$, RMS $= \sqrt{\overline{\Delta T^2}} = \sqrt{\mathrm{bias}^2 + \mathrm{SD}^2}$; wind vector RMS $= \sqrt{\overline{|\vec{V}_a - \vec{V}_s|^2}}$.

Written with the assistance of Claude (Anthropic). Screenshots: MODE-S Wind, dark theme, with the database and the Jokioinen soundings (FMI open data, CC BY 4.0) of 5–21 October 2026 (Figs. 10–12 from the database of 20 October, Figs. 13–16 and Section 7 from the database of 21 October, shown as at 21 October 12 UTC; Figs. 17–19 from the database of 21 October 14:10 UTC; in Section 7 the map tiles are replaced by the coastline and border outlines of the project's overlays). The surface values in Fig. 1 are those of the METAR stored with the hourly profile of 19 October 07 UTC; the page was shown as at 19 October 2026 08:19 UTC.